

Time-varying Informed and Uninformed Trading Activities ¹

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Abstract

We develop a framework to investigate time-varying informed and uninformed trading activities and the relationship between them. Informed traders may match the level of the uninformed arrival rate with certain probability so as to make better use of the camouflage provided by the uninformed trades.

Our empirical estimation of 40 NYSE stocks shows that the buy and sell arrival rates of the uninformed traders are different and time-varying. Uninformed traders tend to adopt contrarian strategy in reaction to high prior stock returns, but employ momentum strategy in reaction to high prior market returns. The estimated probability of matching response ranges from 0.72 to 0.98, reflecting that informed traders are taking good advantage of the camouflage provided by the uninformed traders. The estimated time-varying probability of informed trading is a good predictor for various measures of bid-ask spreads, and is a better measure of information asymmetry than several existing measures.

1 Introduction

A crucial topic in market microstructure is the relationship between informed and uninformed trading activities. From the early observations of Bagehot (1971) to the theoretical work of Kyle (1985) and Easley and O'Hara (1987), researchers generally agree that informed traders exploit their informational advantage and trade optimally to profit from uninformed investors.

The interaction between asymmetrically informed traders, however, has been mostly investigated in theoretical frameworks. The relatively few empirical studies mostly focused on cross-sectional analysis and used very short samples.¹ Using ten years of transaction data, this paper provides empirical perspectives on the time-varying trading behavior of informed and uninformed traders. Moreover, this paper estimates a measure of time-varying probability of information-based trading, which could be a very useful tool in empirical market microstructure analysis. For example, a recent paper by Llorente, Michaely, Saar and Wang (2002) analyzed the dynamic relation between daily volume and first-order autocorrelation for individual stocks. They found that the variation in the relation is related to the extent of informed trading. Since there was no direct measure of the extent of the informed trading, the authors used noisy proxies for information asymmetry such as bid-ask spread and market capitalization. With the estimated probabilities of information-based trading, their hypothesis could be tested more directly.

Specifically, this paper addresses the following questions: Does uninformed trading change over time? How does it change? Does the probability of information-based trading change over time? If yes, does this probability has more explanatory power than competing measures of information asymmetry? Not surprisingly, we find both the informed and uninformed trading change dramatically over the ten-year period. The uninformed buy and sell arrival rates are different and time-varying. The estimated probabilities of information-based trading are closely related to contemporaneous bid-ask spreads and can predict spreads for the next trading day. As an application of the time-varying probability of information-based trading, we verify the relation between information asymmetry and return autocorrelation, which is the focus of Llorente et al. (2002).

In this paper, we extend the seminal work of Easley, Kiefer, O'Hara and Paperman (1996), which inquired why there is a larger spread observed for less-frequently traded

¹Wang (1993) showed that information asymmetry among investors can increase price volatility. Less informed traders may rationally behave like price chasers, which may in turn increase market volatility. See also Diamond and Verrecchia (1981), Glosten and Milgrom (1985), Kyle (1985), Admati (1991), Easley and O'Hara (1992), and Easley, Kiefer, and O'Hara (1997).

stocks as opposed to the active ones. In the absence of market power, market makers charge a bid-ask spread in stock transactions to recover the losses to informed traders and inventory costs.² In Easley et al. (1996), informed traders submit buy orders upon receiving a positive information signal and submit sell orders upon a negative information signal. Uninformed traders, on the other hand, submit both buy and sell orders regardless of whether information arrives or what type of information arrives. By assuming that orders submitted by informed traders, uninformed buyers and uninformed sellers follow three independent Poisson processes with exogenous, fixed arrival rates, Easley et al. (1996) derived a very useful framework to examine the information content of stock trading. They found that the risk of information-based trading is lower for active stocks than it is for infrequently traded securities. Although high-volume stocks tend to have higher probabilities of information events and higher arrival rates of informed traders, they are more than offset by the higher arrival rates of uninformed traders. From the perspective of the market maker, the less active stocks are riskier since there is a higher probability that any trade comes from an informed trader.³

It seems restrictive, however, to assume a constant arrival rate for uninformed traders since the amount of uninformed trading could change dramatically under different market conditions. For instance, with the advent of Internet trading, hundreds of thousands of small investors traded technology stocks in the late 1990s when the NASDAQ Composite Index reached over 5,000. The subsequent stock market correction brought the index down to 1,200 in mid 2002. Many small investors reduced their trading activities, and the market meltdown in technology stocks forced many former day-traders to quit trading altogether. The time-invariant arrival rates also restricted Easley et al. (1996) to study a short sample covering less than three months of daily trading data, since a longer sample period would make it implausible to assume constant arrival rates.

The level of uninformed buy and sell arrival rates may be affected by momentum

²Bid-ask spread arises naturally due to both inventory (see Smidt (1971), Garman (1976), Zabel (1981), Mendelson (1982), O'Hara and Oldfield (1986), and Madhavan and Smidt (1993)) and asymmetric information concerns (Bagehot (1971), Glosten and Milgrom (1985), and Easley and O'Hara (1987)). Glosten and Harris (1988) decompose the bid-ask spread into two parts, the part due to informational asymmetries, and the remainder, which can be attributed to inventory carry costs, market maker risk aversion, and monopoly rents. Using a maximum likelihood technique, they find that the adverse selection component of the bid-ask spread is not economically significant for small trades, but increases with trade size. See also Hasbrouck (1991ab), Barclay, Litzenberger and Warner (1990), Madhavan and Smidt (1991, 1993), Jones, Kaul, and Lipson (1994), and Madhavan and Sofianos (1998).

³The Easley et al. (1996) framework was extended by Weston (2001) to include a class of discretionary liquidity traders.

(Jegadeesh and Titman (1993, 2001), Chan, Jegadeesh, and Lakonishok (1996), Hong and Stein (1999), Rouwenhorst (1999), and Caginalp, Porter, and Smith (2000)) and contrarian strategies (Lo and MacKinlay (1990), and Lakonishok, Shleifer, and Vishny (1994)). It may also be affected by factors such as investor sentiment (Siegel (1992), and Barberis, Shleifer and Vishny (1998)), overconfidence (Daniel, Hirshleifer, and Subrahmanyam (1998)), loss aversion, and mental accounting (see, for example, Barberis and Huang (2001), and Barberis, Huang, and Santos (2001)). We assume that the arrival rate of uninformed buy orders switch between two levels in a Markov process, with endogenous time-varying transition probabilities. We model the difference between buy and sell arrival rates for uninformed traders to be time-varying and dependent on market variables such as lagged cumulative returns. Our framework can be generalized to more than two states, but we find that the parsimonious assumption of two states is sufficient to capture the main effects of time-varying uninformed trading.

We argue that informed traders closely monitor market movements and can respond rationally to any change in the arrival rate of uninformed traders. The theoretical framework in Glosten and Milgrom (1985) and Easley et al. (1996) assume that traders are chosen probabilistically to submit an order of one share at each session, so informed traders cannot respond to camouflage provided by the uninformed traders. Our framework allows informed traders to match the arrival rate of uninformed investors by assuming that the arrival rate of informed traders also can switch between two levels, with the transition probabilities reflecting informed matching activities (to be defined shortly).

Kyle (1985) and Back (1992) demonstrated elegantly the strategic trading behavior of informed traders in a risk-neutral environment, and the setup of this paper amounts to an empirical test of their theoretical predictions. We can examine how well informed traders use the camouflage provided by uninformed investors. With the endogenously determined arrival rates for both types of traders, we are able to evaluate the evolution of information content in daily stock trading. Easley et al. (1996) performed a cross-sectional study on the relationship between the probability of information content and the daily opening spread. Since we use a much longer sample and allow time-varying probability of information-based trading, we are able to conduct the analysis in both the time series and the cross-sectional dimensions. Moreover, we can study the predictability of the estimated probabilities of information-based trading for various measures of stock spreads.

Our empirical estimation of 40 NYSE stocks shows that the uninformed traders tend to adopt contrarian strategy in reaction to high prior own stock returns, but employ momentum strategy in reaction to high prior market returns. This result is

consistent with the finding of Jegadeesh and Titman (1995) that stocks overreact to stock-specific factors and underreact to market-wide factors. Informed traders seem to take good advantage of the camouflage since the estimated probability of informed matching response ranges from 0.72 to 0.98. The estimated probability of informed trading has predictive power over mean bid-ask spreads in the next trading day, and dominates competing measures of information asymmetry in terms of explanatory power. Moreover, we use the estimated time-varying probability of informed trading series as a proxy for informational asymmetry to analyze its impact on the serial correlation of daily stock returns. Consistent with existing research on the issue with other measures of informational asymmetry such as firm size and the bid-ask spread, we find that higher probability of informed trading is associated with higher return autocorrelation.

The rest of the paper is organized as follows. Section 2 extends the framework in Easley et al. (1996) by allowing the uninformed arrival rates to be time-varying. Informed traders may match the level of the uninformed arrival rate with certain probability so as to make better use of the camouflage provided by uninformed traders. In Section 3, we discuss the sample selection and other data-related issues. Section 4 analyzes the maximum-likelihood estimation results and the cross-sectional variation of matching response by informed traders. In Section 5, we examine the predictability of the estimated probabilities of informed trading for various measures of spreads, and the effect of information asymmetry on return autocorrelation. Section 6 concludes the paper. Some technical details are provided in the Appendix.

2 The Model

There is one risky asset and one risk-free asset (as the *numeraire*) in the market. The risk-free rate is set to be zero for simplicity. At the beginning of each trading day $t \in [1, T]$, in which time is continuously indexed by $0 < i < 1$, nature decides whether to release a news event concerning the value of the risky asset to informed traders. The probability of a news event occurring is α , and when the news event arrives it may be bad news with probability δ or good news with probability $1 - \delta$. Hence the prior probabilities of having a negative, positive, and no news event are $p(-) \equiv \alpha\delta$, $p(+) \equiv \alpha(1 - \delta)$ and $p(\times) \equiv 1 - \alpha$, respectively. The buy and sell trades from the uninformed traders as well as the one-sided trades (buy at good news and sell at bad news) from the informed traders are assumed to follow three mutually independent Poisson processes.

2.1 Two-State Markov Switching for the Uninformed Arrival

On each trading day, the uninformed arrival rate is assumed to follow a two-state Markov switching process, with the time-varying transition probabilities governed by factors such as prior stock return and prior market return. The uninformed arrival rate can be at the high level ε_t^h or at the low level ε_t^l on trading day t . Note that the high and low levels of the arrival rate are constant for the entire sample, despite the time subscript they carry for notational clarity.

Define the time-varying transition probabilities for the uninformed traders as

$$\pi_t \equiv \begin{bmatrix} \pi_t^{**} & 1 - \pi_t^{**} \\ 1 - \pi_t^* & \pi_t^* \end{bmatrix},$$

where

$$\pi_t^* \equiv \Pr(\varepsilon_t^l | \varepsilon_{t-1}^l) = f(\beta_l' \mathbf{z}_t), \text{ and } \pi_t^{**} \equiv \Pr(\varepsilon_t^h | \varepsilon_{t-1}^h) = f(\beta_h' \mathbf{z}_t). \quad (1)$$

In the definitions above, π_t^* is the probability of the uninformed arrival rate transiting from the low level at period $t - 1$ to the low level at period t , and π_t^{**} is the probability transiting from the high level at period $t - 1$ to the high level at period t . $\mathbf{z}_t$ is a vector of instruments that are observable at the end of period $t - 1$. If we use a vector of ones as the only instrument, then the model yields constant transition probabilities. Non-constant instruments, such as cumulative asset return and cumulative market return (value-weighted NYSE/AMEX/NASDAQ returns) in the previous 20-trading-day period, allow the modeling of time-varying transition probabilities. We use the logistic transformation $f(\cdot)$ to ensure the transition probabilities have appropriate values. β_h and β_l are the parameters associated with the instruments.

Uninformed investors may base their buying and selling activities on publicly observable information, such as past stock and market returns. We allow the uninformed buy arrival rate to be different from the uninformed sell arrival rate. However, the modeling of two separate uninformed arrival rates, each with its own Markov switching process, forces us to track the interaction between them. As a consequence, it significantly reduces the tractability of the empirical model and renders the estimated results very difficult to interpret. We take a modeling compromise, and allow only the uninformed buy arrival rate ($\varepsilon_t^{h,B}$ for the high state and $\varepsilon_t^{l,B}$ for the low state) to follow the aforementioned two-state Markov switching process. The uninformed sell arrival rate ($\varepsilon_t^{h,S}$ for the high state and $\varepsilon_t^{l,S}$ for the low state) switches into the same state as the uninformed buy arrival rates, but the level of two arrival

rates may differ. In particular, we assume that

$$\varepsilon_t^{h,S} = \varepsilon_t^{h,B} \exp(\gamma_h' \mathbf{z}_t), \text{ and } \varepsilon_t^{l,S} = \varepsilon_t^{l,B} \exp(\gamma_l' \mathbf{z}_t). \quad (2)$$

Note that we are using the same set of instruments $\mathbf{z}_t$ here as in the context of modeling time-varying transition probabilities. The assumption that the uninformed buy and sell arrival rates will switch to the same state simultaneously, either high or low, is not as restrictive as it seems. Researchers on the trading behavior of small investors find that retail trading activities increase after up-markets (Odean (1999) and Grinblatt and Keloharju (2001)). They are more likely to sell stocks to realize gains and buy stocks because of overconfidence. During down-markets, retail investors are reluctant to sell to realize losses due to loss aversion, and they buy less due to a lack of interest or attention (Odean (1998) and Barber and Odean (2003)). Our simple structure also brings forth multiple benefits: achieving empirical tractability and ease of interpretation, and allowing us to focus on the extent of (percentage) difference between the uninformed buy and sell arrival rates in response to changes in market fundamentals. We call this difference “uninformed arrival rate differential” henceforth.

Since the amount of uninformed trading can vary dramatically from time to time, our switching model should improve the Easley et al. (1996) model that has constant arrival rates. A recent study by Easley, Engle, O’Hara, and Wu (2002) also relaxed the assumption of fixed arrival rates in Easley et al. (1996) by using two GARCH specifications. We use a Markov-switching model instead to investigate how the behavior of the uninformed investors may affect the strategic trading behavior of the informed traders. Our use of the Markov switching model is motivated by the observation that uninformed traders become more or less interested in trading depending on past market performance. Our focus is the interaction between informed and uninformed trading, and a Markov switching process makes it easy to estimate and interpret the interaction. In the next sub-section, we extend the model further to examine how well informed traders are making use of the camouflage from uninformed traders by allowing informed traders to engage in level-matching activities. The Markov-switching model makes it a straightforward and easy task to interpret the matching probabilities. Our framework also can easily accommodate the use of instruments in the modeling of transition probabilities, and it provides a very intuitive interpretation to the estimated relationship.

2.2 Level-matching Activities by the Informed Traders

The informed traders may adjust their trading activities according to the amount of camouflage provided by uninformed traders (Kyle (1985)). In the market microstructure model of Glosten and Milgrom (1985) and Easley et al. (1996) traders can not trade more than one unit of the asset per period. Without this assumption, informed traders will buy (sell) an infinite amount upon the arrival of good (bad) news. So we preserve the assumption of unit trade in this paper and attempt to model the strategic behavior of informed traders by studying the fraction of time when both types of traders make synchronized moves in terms of the state level of arrival rates. That is, we allow the informed arrival rate on day t to take on a high level μ_t^h (or a low level μ_t^l), corresponding to the high level ε_t^h (or low level ε_t^l) of the uninformed arrival rates, a situation we call “level-matching activities” in this paper. The empirically estimated probability of the matching response should shed some light on how well the informed traders are making use of the camouflage.

There are four possible combinations of the uninformed and the informed arrival rates in our model, denoted as $s_t^a \equiv (\varepsilon_t^h, \mu_t^h)$, $s_t^b \equiv (\varepsilon_t^h, \mu_t^l)$, $s_t^c \equiv (\varepsilon_t^l, \mu_t^h)$, and $s_t^d \equiv (\varepsilon_t^l, \mu_t^l)$.⁴ Clearly, the two states s_t^a and s_t^d are the matching states and the rest are mis-matching states. Now we have a modeling choice in terms of tracking the interaction between the activities of the informed traders and those of the uninformed traders. Though it is not impossible to impose a Markov-switching structure on the four composite states, there is no easy way to extract intuition regarding the informed-uninformed interaction from the complex four-state transition matrix. This is a problem similar to the one we face when modeling the relationship between the uninformed buy arrival rate and the uninformed sell arrival rate. Yet we must not resort to the same solution as before because an assumption that the informed traders move in a fully synchronized fashion with the state level of the uninformed traders completely destroys our goal of uncovering the extent of strategic movement by the informed traders. Given that a hard-wired complete synchronization between the informed and the uninformed traders is out of the question, we decide to impose some structural restrictions on the four-state transition matrix such that the informed traders would match the level of their arrival rate with the level of the uninformed arrival rate only at a fraction of the time. We hope to study the estimated value of this fraction for matching activities, a parameter we call “probability of matching response,” to draw useful inference on the informed traders’ strategic

⁴Note that we use the abbreviated notation for the uninformed arrival rates, without distinguishing the uninformed buy and sell arrival rates, due to the assumption of adaptive movement in equation (2). We use the abbreviated notation whenever it is not likely to cause confusion.

behavior, while allowing the uninformed arrival rate to switch freely according to the two-level Markov process described earlier.

Provided that the informed traders know the level of the uninformed arrival rate at the beginning of each trading day, they can make a decision on how to exploit the information advantage they have. In particular, we define the probability of matching response as

$$\Pr(\mu_t^h | \varepsilon_t^h) = \Pr(\mu_t^l | \varepsilon_t^l) \equiv \rho. \quad (3)$$

Moreover, the time-varying probability of transiting from state s_{t-1}^a to s_t^a is equal to:

$$\Pr(s_t^a | s_{t-1}^a) = \Pr(\varepsilon_t^h, \mu_t^h | \varepsilon_{t-1}^h, \mu_{t-1}^h) = \Pr(\varepsilon_t^h | \varepsilon_{t-1}^h) \cdot \Pr(\mu_t^h | \varepsilon_t^h, \varepsilon_{t-1}^h, \mu_{t-1}^h) = \pi_t^{**} \cdot \rho. \quad (4)$$

The first equal sign follows the definition of composite states. The second equal sign applies because the uninformed traders don't have knowledge about current level of the informed arrival rate and we assume that the uninformed arrival rates switch freely according to the two-state Markov process defined earlier. Implicitly, it is assumed that the trading behavior of the uninformed investors is completely governed by whatever information variables we use for the uninformed transition probabilities and the uninformed arrival rate differential. The third equal sign follows from the definition of matching response probability as well as the assumption that the informed traders pay no attention to arrival rates in the previous trading day.

To get a better grasp of the parameter ρ , we may analyze its inherent limitations first. First, its existence hinges upon the independence assumption about the uninformed investors. If the uninformed investors have knowledge of the informed arrival rate and use this knowledge to decide their choice of arrival rate level and arrival rate differential, then the second equal sign in the equation above does not go through and the so-called matching response probability cannot stand. Perhaps we can get around this problem by re-classifying the trader types. That is, any uninformed traders who are sophisticated enough to infer the informed arrival rate and actively use that inference for trading decisions can be re-classified as informed traders⁵.

Second, given that the parameter ρ is a statistical construct, arising from the manual decomposition of the transition probability of composite states, we are still not very sure of its path-independent nature and its constancy. The path-independent assumption on the matching response by the informed traders can be partially motivated by the information advantage they possess as the informed traders trade

⁵The argument of reclassification of trader types is stated from a theoretical perspective, and we do not attempt to empirically classify each trade into the informed or the uninformed categories. The distinction of the arrival rates for these two types of traders is achieved via the sample likelihood function (to be discussed later), utilizing the fact that the informed traders submit trades on only one side based upon the type of private signal they receive.

based upon private signals that may arrive on each trading day. So it is reasonable to assume that the informed investors do not care about their own arrival rate in the previous trading day. But given that the informed investors care about the existing level of the uninformed arrival rates, which is related to the uninformed arrival rate in the previous period, why would the informed investors ignore the uninformed arrival rate in the previous trading day? Moreover, what makes us to believe that the probability of matching response as defined is constant throughout the sample period? To address these two questions, we have estimated more complex versions of the model on a subset of our sample stocks. In particular, we allow the matching response to be dependent upon the uninformed arrival rate in the previous period, such as $\Pr(\mu_t^h | \varepsilon_t^h, \varepsilon_{t-1}^h) \neq \Pr(\mu_t^h | \varepsilon_t^h, \varepsilon_{t-1}^l)$, and even make the probability of matching response vary over time. We find the path-dependent matching response probabilities are not significantly different from each other, and the sample standard deviation of the time-series of matching response is very small. So we conclude these two assumptions in our current model are reasonable.

In sum, the probability of matching response as proposed is an imperfect way of describing the interaction between the informed traders and the uninformed. But it is a reasonable measure because the assumptions we made for this statistical construct are not farfetched. After all, an empirical assumption can only be tested in an empirical way, and we are eager to see if the estimated probability of matching response by the informed traders is statistically significant and if the magnitude of this estimate is as large as suggested by theory. If ρ is close to 1, then we have evidence that the informed traders are using the camouflage from the uninformed traders. Admati and Pfleiderer (1988) present a theory in which concentrated trading patterns arise endogenously as a result of the strategic behavior of (uninformed) noise traders and the informed traders. Although their theory is used mainly to explain intra-day trading patterns, our model presents an empirical test of their results for inter-day patterns (see also Back and Pedersen (1998)).

Finally, the time-varying nature of the uninformed transition probabilities and the uninformed arrival rate differential allows us to deliver a time series of a variable measuring the probability of the informed trading (TPIN). That is, the probability that each trade on day t is information-based can be expressed as

$$\text{TPIN}_t = \frac{\alpha \mu_t^e}{\varepsilon_t^{e,B} + \varepsilon_t^{e,S} + \alpha \mu_t^e}. \quad (5)$$

In this definition, α is the probability of news arrival; μ_t^e is the expected arrival rate for the informed traders on trading day t ; $\varepsilon_t^{e,B}$ is the expected buy arrival rate for the uninformed traders on trading day t ; and $\varepsilon_t^{e,S}$ is the expected sell arrival rate

for the uninformed traders on trading day t . This definition is identical in spirit to the constant version of PIN defined in Easley et al. (1996), and it has an intuitive interpretation as the fraction of informed trades among all trades. The short-hand TPIN is used in this paper to signify its time-varying nature, as opposed to the constant PIN. The technical details about the composite transition matrix as well as the definitions of the expected arrival rates are provided in the Appendix.

2.3 Trading Process and Estimation Methodology

The basic structure of the trading process on a typical trading day is depicted in Figure 1. Nodes above the dotted line occur only once per trading day, and nodes on the dotted line are repeated many times within each trading day, following independent Poisson processes by assumption. Depending upon the previous level $\varepsilon_{t-1}^?$ of the uninformed arrival, the uninformed arrival may transit into the high level with probability $p(\varepsilon_t^h|\varepsilon_{t-1}^?)$ or the low level with probability $p(\varepsilon_t^l|\varepsilon_{t-1}^?)$. The informed traders engage in level-matching activities with a constant probability ρ . There may be no news arrival with probability $p(\times)$, good news arrival with probability $p(+)$, or bad news with probability $p(-)$. The last row of numbers is the sum of the expected arrival rate of the informed and the uninformed traders under each scenario, with notational abbreviation for the uninformed arrival rates.

Figure 1 forms the basis of our empirical estimation. In the Appendix, we construct the likelihood function in a computation-efficient way. Given the likelihood function, all the parameters can be estimated using an optimizer such as Maxlik in GAUSS, the statistical package we use for this study. Due to the one-sided nature of the informed trades, the sample log-likelihood function helps us distinguish the informed arrival rates from the uninformed arrival rates. Although it is not possible to empirically identify the particular composite state for each trading day, we can nevertheless make probabilistic statements about the informed and the uninformed arrival rates for each trading day for the entire sample.

3 Data Construction

Our sample of firms was selected from 611 common shares listed on the NYSE (Share Code 10 or 11) with complete monthly returns data from the Center for Research in Securities Prices (CRSP) database and trading data from the New York Stock Exchange Trade and Quote (NYSE TAQ) database. These stocks did not change their respective ticker symbol during the ten-year period between 1993 and 2002. We sort these stocks into 10 deciles according to the monthly average turnover ratio

(the number of shares traded to the shares outstanding) and randomly pick 10 stocks from each of the 2nd, 4th, 6th and 8th turnover deciles. We use turnover as a sample selection criterion because Easley et al. (1996) demonstrates that low volume is indicative of high information asymmetry and turnover has been used as a measure of volume in the literature (see, for example, Blume, Easley and O'Hara (1994), Lee and Swaminathan (2000), and Lo and Wang (2000)).

Before matching trades with quotes, we collapse the closely adjacent (within five seconds) trades that were executed at the same price without intervening revision of quotes. This is done to mitigate the problem of misclassifying as separate trades a large trade on one side (buy or sell) that involved multiple participants. We also allow for a systematic delay of five seconds before each submitted quote was time-stamped. We treat as relevant only the most recent quote that was at least five seconds old relative to the recorded trade time.

We follow the methodology proposed by Lee and Ready (1991) in determining if a trade is buyer-initiated or seller-initiated. Trades with transaction prices above the midpoint of the relevant bid and ask are called buy trades, and trades with transaction prices below the midpoint are called sell trades. Trades with a price at the midpoint of bid and ask are classified with a "tick test," i.e., trades with a price higher (or lower) than the most recent trade with a different price are called buy (or sell) trades.

Table 1 reports some summary statistics about the stocks in our sample. The summary statistics show that there are a lot of variations in firm characteristics, both within each turnover decile and across the turnover deciles. Stocks with higher daily mean turnover tend to have higher stock prices and more transactions (both buy trades and sell trades). The mean stock price ranges from \$4.19 to \$63.82, so our sample does not include penny stocks. There is no clear pattern across turnover deciles for the mean cumulative prior-20-trading-day stock return, which ranges from -110.20 basis points to 174.07 basis points. The average mean turnover for stocks in the 2nd turnover decile is 16.36 basis points per day or 41% per year, and the average mean turnover for stocks in the 8th turnover decile is 44.27 basis points per day or 112% per year. The variation in the standard errors for the variables presented in Table 1 is similar to the variation in the means of these variables.

4 Estimation Results

In this section, we discuss properties of the estimation for each stock in our sample as well as the implied time series related to the information content of stock trading. We defer to the next section some applications of the estimated probability of informed

trading.

4.1 Maximum Likelihood Estimates

Table 2 presents the maximum likelihood estimates for all the constant parameters in our model including the probability of news arrival, the probability of arrived news being negative, different levels of the informed and the uninformed arrival rates, and the probability of matching response. These constant parameters are estimated using the entire sample for each stock, and they are highly statistically significant at the 1% level⁶. There is large variation for the estimated parameters across stocks. For example, the probability of news arrival ranges between 0.2846 for stock ticker AGL and 0.5151 for stock ticker LUB. The high level of the uninformed buy arrival rate can be as high as 101 trades per day for stock ticker OII, or as low as 9 trades for stock ticker SL. The high level of informed arrival rate varies between 92 trades for stock ticker KMT and 16 trades for stock ticker SL. We do not report the levels of the uninformed sell arrival rates, because we model the sell arrival rates based upon the corresponding state level of the uninformed buy arrival rates, while allowing the difference between the uninformed buy and sell arrival rates to be time-varying and responsive to market fundamentals. The constant probability of matching response by the informed traders varies between 0.7187 for stock ticker NEV and 0.9826 for stock ticker AGL. It is perhaps worth noting that all estimated probability of matching response is statistically different from 0 and 0.5. The matching response is very close to 1, a level indicative of highly synchronized movement between the uninformed and the informed traders. Therefore, informed traders do make good use of the camouflage provided by uninformed trades, as suggested by Kyle (1985), Admati and Pfleiderer (1988), and Back and Pedersen (1998).

The time-varying nature of our model makes it natural to deliver a few important time series related to information content of stock trading activities. In Table 3, we present the sample mean and standard deviation for the expected uninformed buy arrival rate, the expected uninformed sell arrival rate, the expected informed arrival rate, and the expected probability of informed trading (TPIN). There is a lot of variation across different stocks for different arrival rates, and we typically see the sell arrival rate at a level slightly higher than the buy arrival rate for the uninformed traders. The sample mean TPIN ranges from 0.1823 for stock ticker AGL to 0.3007 for stock ticker RTI. We defer to a later section discussions on the link between

⁶We omitted reporting the standard errors for these parameter estimates in order to conserve space, and the complete estimation results are available upon request.

cross-sectional variation of sample TPIN and firm characteristics.

4.2 Influence of Market Fundamentals

Details about the estimates related to information variables, including cumulative prior-20-trading-day stock return (CDR) and cumulative prior-20-trading-day market return (CMR), are presented in Figures 2 and 3. A constant also is included as part of the instruments vector for the uninformed transition probabilities (UT) and for the uninformed arrival rate differential (UD), but we do not report the estimated constants in order to save space. We made a decision to graphically present the estimated coefficients associated with both information variables CDR and CMR because the numeric version simply contains too many numbers. In order to deliver a clear and intuitive pattern, we do not show any estimated coefficient that is not significantly different from 0 at the 5% level. All insignificant estimates are replaced with zero.

The first two rows of Figure 2 depict the influence of prior stock returns on the uninformed transition probabilities (UT), and the last two rows of the figure graphs the influence of prior stock returns on the uninformed arrival rate differential (UD). Each row of the figure contains four individual plots with a common scale, one for each turnover decile, and stocks within each turnover decile are presented in alphabetic order by tickers. The first impression of Figure 2 is that we see a lot more bars (standing for estimates significant at the 5% level) than dashed segments (standing for insignificant estimates). Therefore, the uninformed traders are quite responsive to prior stock returns that are part of the instrument vector $\mathbf{z}_t$.

In reaction to prior stock returns, the uninformed traders in the model need to decide trade more or less frequently, i.e., to decide which state level of arrival rate to follow. Given high prior stock returns, this response can be uniformly positive (i.e., choosing the high state of arrival rate regardless of the current state level), uniformly negative (i.e., choosing the low state of arrival rate regardless of the current state level), or path-dependent (i.e., switching the state level of arrival rate in the next period based upon the existing state level). Specifically, a uniformly positive response on the state level of the uninformed arrival rate is represented by a positive coefficient for staying at the high state arrival rate, denoted by label UT(H2H) in the figure, and a negative coefficient for staying at the low state arrival rate, denoted by label UT(L2L). A negative coefficient for UT(H2H) coupled with a positive coefficient for UT(L2L) is indicative of a uniformly negative response to prior stock returns. A path-dependent response to prior stock returns is represented by the same signs on UT(H2H) and UT(L2L).

Given the current state of high uninformed arrival rate, the uninformed traders on 14 stocks are more likely to make the switch to the low state, and the uninformed traders on eight stocks are more likely to choose staying at the high level, in reaction to high prior stock returns. Given the current state of low uninformed arrival rate, the uninformed traders on 21 stocks are more likely to make the switch to the high level, and the uninformed traders on four stocks are more likely to choose staying at the low level. The pattern of predominantly negative signs for both UT(H2H) and UT(L2L) suggests that the uninformed traders react differently to high prior stock returns, taking a path-dependent response. Specifically, they would switch from the current state level as long as high prior stock returns are observed.

Prior stock returns may also affect another decision by the uninformed traders, i.e., to submit more buy or sell orders. They may follow either a contrarian strategy (i.e., sell after high prior stock returns) or a momentum strategy (i.e., buy after high prior stock returns). The use of the contrarian strategy should be reflected by a positive impact on the uninformed arrival differential, which is defined as the percentage difference between sell arrival rate and buy arrival rate at the particular state level of the uninformed arrival rate. For example, if the uninformed traders are taking the contrarian strategy when they are at the high level of arrival, we should see positive coefficient on the uninformed arrival differential at the high level, denoted by label UD(High) in the figure. Similarly, a positive coefficient on UD(Low) is considered indicative of the uninformed traders taking the contrarian strategy at the low level of arrival. The use of the momentum strategy is associated with negative signs on UD(High) or UD(Low).

From Figure 2, we see that in reaction to high prior stock returns, the uninformed traders on 19 stocks will sell significantly more than buy when the existing uninformed arrival rate is high, as indicated by a significantly positive coefficient on UD(High). The uninformed traders on 11 stocks significantly buy more than sell as indicated by a significantly negative coefficient on UD(High). That is, we see more use of the contrarian strategy than the momentum strategy by the uninformed traders when the amount of uninformed trading is high. When the existing uninformed arrival rate is low, we see that the uninformed traders on 16 stocks significantly sell more than buy, as indicated by a significantly positive coefficient on UD(Low). The uninformed traders on only eight stocks buy more than sell, as indicated by a significantly negative coefficient on UD(Low). Again, we see more use of the contrarian strategy than the momentum strategy when the amount of uninformed trading is low. Therefore, the evidence from our sample of stocks suggests that uninformed traders tend to adopt a contrarian strategy in reaction to high past stock returns. It is interesting to note that Lo and MacKinlay (1990) found it profitable to employ a

contrarian strategy in very short horizon such as one month.

In Figure 3, we report the results on the response of the uninformed traders to past market returns. Since the market is common to all stocks, we can make direct comparison across stocks. For the influence of prior market returns on the uninformed transition probabilities, we see that the uninformed traders prefer switching into or staying at the low level of arrival rate. That is, 11 stocks have a significantly negative coefficient on $UT(H2H)$ as opposed to five stocks with significantly positive coefficient on $UT(H2H)$, and 12 stocks have a significantly positive coefficient on $UT(L2L)$ as opposed to three stocks with significantly negative coefficient on $UT(L2L)$. Given that many stocks have an insignificant coefficient associated with prior market returns, we should avoid over-interpreting the impact of prior market returns on the uninformed transition probabilities.

The more interesting results lie in the third and fourth rows of Figure 3. We see 20 stocks with significantly negative coefficients on $UD(High)$ and four stocks with significantly positive coefficients on $UD(High)$, indicating that the uninformed traders are predominantly using momentum strategy in reaction to high prior market returns when the uninformed trading activities were high in the last period. When the uninformed trading activities were low, the uninformed traders still employ the momentum strategy more often since 15 stocks have significantly negative coefficients on $UD(Low)$ as opposed to eight stocks have significantly positive coefficients on $UD(Low)$. Comparing the magnitude of impact on $UD(High)$ to that on $UD(Low)$, we find a much stronger momentum effect when the existing uninformed traders are active. Comparing the magnitude of impact on stocks across different turnover deciles, we find a stronger momentum effect on the lower turnover deciles. The broad message from Figure 3 is that the uninformed traders tend to adopt the momentum strategy in reaction to high prior market returns, and submit more buy orders than sell orders.

Various studies have shown that different types of investors have very different trading styles. For example, Barber and Odean (2003) find that individual investors tend to submit more buy orders than sell orders in reaction to extreme positive returns in the past. Choe, Kho and Stulz (1999) find strong daily evidence that Korean and foreign institutional investors use “positive feedback and herding” trading strategies. Moreover, in a study on the daily and intra-day trading of NASDAQ 100 stocks, Griffin, Harris and Topaloglu (2003) find the presence of trend chasing behavior by institutional investors as well as evidence supporting contrarian strategy adopted by individual investors. Therefore, it is not surprising to expect different trading behavior on the aggregate level for different stocks, for they potentially have very different institutional ownership structure. Our empirical model does not make

a distinction between institutional and individual investors, so the trading style by our uninformed investors may vary across different stocks.

The number of insignificant estimates (represented by a dash in Figure 2 and 3) makes one wonder if it is reasonable after all to assume that the uninformed buy and sell arrival rates differ in a time-varying fashion. Note that, however, very few stocks have insignificant UD parameters for both instruments, and no stock has its UD and UT parameters insignificant simultaneously. We further conduct likelihood ratio tests of the model against two alternative models: one with constant transition probabilities and constant buy-sell differentials, and another with time-varying transition probabilities but constant buy-sell differentials. Table 4 shows that both of these alternative models are rejected for every stock in the sample at the 1% significance level. We conclude that there is strong evidence in support of time-varying uninformed buy and sell arrival rates and fundamentals such as prior stock/market returns do play a role.

In sum, we find that uninformed investors tend to adopt the contrarian strategy in reaction to high prior own-stock returns and adopt the momentum strategy in reaction to high prior market returns. Researchers such as Jegadeesh (1990) and Lehmann (1990) find that short-term own-stock contrarian strategies yield abnormal returns. This short-term reversal may reflect corrections for “over-reactions” in stock prices or inefficiency in the market for liquidity around large price changes. Jegadeesh and Titman (1995) find that stock prices overreact to firm specific factors and underreact to market-wide factors. Although they use a framework totally different from ours, the behavior of uninformed investors implied by our model is nevertheless consistent with their findings.

4.3 Cross-sectional Variation of Matching Response

We note with interest that the estimated probability of matching response in Table 2 shows considerable variation across stocks. In particular, the estimated probability of matching response is negatively correlated with the estimated probability of informed trading among our sample of 40 stocks, with a correlation coefficient -0.39 . Consistent with the finding in Easley et al. (1996) that less frequently traded stocks have more asymmetric information, we obtain a cross-sectional correlation of -0.31 between the estimated probability of informed trading and the dollar volume. To better understand the relationship between the informed matching activities and firm characteristics, we design a cross-sectional regression as follows,

$$\rho_i = \beta_0 + \beta_1 \cdot \text{INVOL}_i + \beta_2 \cdot \text{UNVOL}_i + \varepsilon_i, \text{ where } i = 1, \dots, 40. \quad (6)$$

In this equation, the independent variable ρ is the estimated probability of matching response, and the informed dollar volume INVOL is the product of sample average probability of informed trading and average dollar volume. The uninformed dollar volume UNVOL is the product of implied probability of uninformed trading, i.e., one minus sample average probability of informed trading, and the average dollar volume.

The goal of this empirical design is to find out the partial effect of the uninformed trading activities on the matching response by the informed traders, after controlling for the competition among the informed traders. We expect to find a positive and significant coefficient associated with the uninformed dollar volume, if the informed traders were indeed motivated to make good use of the camouflage from the uninformed activities. The highly significant results for this regression, shown in Table 5, suggest that the extent of matching activities by the informed traders is higher among stocks with large uninformed dollar volume, while the increased presence of the informed traders undercuts the efforts of matching. In the absence of a rigorous model uncovering the exact determinants of the matching response, it seems a plausible conjecture that the higher competition among informed traders (represented by either a large number of informed traders or higher order flows by the same number of informed traders) makes it harder for all informed traders as a group to match with the uninformed trading activities in a synchronized fashion. On the other hand, increased level of uninformed trading provides more camouflage for informed traders to undertake matching response activities.

5 Applications

In this section, we focus on a few direct applications of the extended model. We first explore the contribution of information asymmetry to various measures of stock spreads by providing evidence from both the cross-sectional and the time-series perspectives. We then examine the predictive power of the estimated probability of informed trading (TPINs) for mean spreads in the next period, along with competing measures of information asymmetry. Finally, we also use the estimated TPINs as a direct measure of information asymmetry and analyze its impact on the persistence of stock returns.

5.1 The Explanatory Power of PINs

To examine the extent to which information asymmetry in stock trading activities may affect various measures of stock spreads, we run a fixed-effect panel regression

that is consistent with the spirit of Easley et al. (1996),

$$\begin{aligned}\Sigma_{i,t} = & \beta_{0,i} + \beta_1 \cdot \text{VTPIN}_{i,t} + \beta_2 \cdot \text{VPIN}_{i,t} + \beta_3 \cdot \text{VOL}_{i,t} + \beta_4 \cdot \delta_2(t) + \beta_5 \cdot \delta_3(t) \\ & + \beta_6 \cdot \delta_2(t) \cdot \text{VTPIN}_{i,t} + \beta_7 \cdot \delta_2(t) \cdot \text{VPIN}_{i,t} + \beta_8 \cdot \delta_3(t) \cdot \text{VTPIN}_{i,t} \\ & + \beta_9 \cdot \delta_3(t) \cdot \text{VPIN}_{i,t} + \eta_{i,t},\end{aligned}\tag{7}$$

where $i = 1, \dots, 40$. The dependent variable Σ is the stock spread of different types extracted from the NYSE TAQ database. The independent variable VPIN is the product of stock price and probability of informed trading PIN estimated from the Easley et al. (1996) framework. Note that a constant PIN is estimated for each quarter of trading data and assigned for that particular quarter. VOL is the dollar volume defined as the product of stock price and share volume (in thousand shares). VTPIN is the product of stock price and probability of informed trading TPIN estimated from our time-varying model. $\delta_2(t)$ is a dummy variable that takes value 1 for the period with tick size being $\frac{1}{16}$ on the New York Stock Exchange (i.e., between June 24, 1997 and January 28, 2001) and $\delta_3(t)$ is a dummy variable that takes value 1 for the period of post-decimalization on the New York Stock Exchange (i.e., on and after January 29, 2001). Empirical studies such as Angel (1997) shows that the minimum tick size has significant impact on the size of the bid-ask spreads.

One important departure from Easley et al. (1996) is that we run a fixed-effect panel regression as opposed to the pure cross-sectional regression in their work. The time series dimension should enable us to discern the contemporaneous contribution of information asymmetry to various measures of spread. We use the VTPIN measure from our extended model along with the original VPIN measure so that we can see if one measure is better than the other.

According to Easley et al. (1996), we expect to see a positive slope on the measure of information asymmetry and a negative slope on the dollar volume VOL. The rationale is that market makers should quote higher spread to offset higher losses to informed trades, and frequently traded stocks command lower spread due to a lesser extent of information asymmetry. Although either of the two measures, VPIN and VTPIN, should enter the regression positively alone, we do not have a prior on how the pattern of signs will play out when both of them enter the regression. One possibility is that one measure may completely subsume the other in explaining spread, so that we see a significant positive coefficient for the dominant measure and an insignificant one for the other. Another possibility is that VPIN and VTPIN measure different dimensions of information asymmetry (i.e., neither measure is a sufficient statistic in capturing information asymmetry), then they may appear positive jointly.

We add two tick size dummy variables, δ_2 and δ_3 , and interactions between the tick size dummies and the information asymmetry measures in the regression above to control for the regime changes in tick size. On June 24, 1997, the tick size on the NYSE has changed from $\frac{1}{8}$ to $\frac{1}{16}$, and after January 29, 2001 the tick size on the NYSE was further reduced to $\frac{1}{100}$, also known as the decimalization. We expect that the reduction of tick size leads to lower spreads and that the post-decimalization period ($\delta_2 = 0$, $\delta_3 = 1$) should exhibit an even stronger reduction in spreads than does the $\frac{1}{16}$ period ($\delta_2 = 1$, $\delta_3 = 0$) relative to the benchmark period with tick size $\frac{1}{8}$ ($\delta_2 = 0$, $\delta_3 = 0$). Put in another way, we expect negative coefficients associated with both δ_2 and δ_3 , and the absolute magnitude of the coefficient for δ_3 should be larger than that for δ_2 .

It is plausible to infer that with tick size reduction liquidity providers face more competition with each other, and thus in the regime with smaller tick size market makers face more constraints in raising the spread in order to recoup the loss to informed traders. That is, market makers may still increase the spread when they face more informed trading, yet they are not able to increase it as much in the smaller tick size regimes as they do in the large tick size regime. We can make two empirical predictions. One is that the coefficient associated with the interaction term between the tick size dummy and the proxy for information asymmetry should be negative, relative to the benchmark case with the largest tick size ($\frac{1}{8}$ in our case). Second, the absolute magnitude of this coefficient should be larger for the post-decimalization period than for the $\frac{1}{16}$ tick size period.

Panel (A) of Table 6 presents the fixed-effect regression results using all available daily observations between January 1993 and December 2002. The signs and magnitude of coefficients, β_4 and β_5 for δ_2 and δ_3 , respectively, conform to the prediction nicely. They are highly significant and negative, with $|\beta_5| > |\beta_4|$, for each of the three versions of spread used as the dependent variable. VOL has the predicted negative sign, and is significant for both daily closing and daily mean spread. Whereas VPIN is significant and positive in explaining the daily closing and daily mean spread, its explanatory power for the opening spread is zero. In contrast, VTPIN is highly significant and positive in all three measures of spread, and the size of the coefficients are much larger than that for VPIN. Because the sample mean of VTPIN is larger than that of VPIN (4.5 versus 3.5) the overall explanatory power of VTPIN dominates that of VPIN. As expected, the interaction terms between VTPIN and the two tick size dummies are significantly negative in five out of six cases. Furthermore, the interaction term with δ_3 is indeed larger in absolute value than that with δ_2 . Hence the results on VTPIN are strong and are consistent with empirical predictions. In contrast, VPIN has significant and positive coefficients for its interaction with tick

size dummies. This is counter-intuitive since it implies that VPIN predicts higher influence of information asymmetry for the regime with smaller tick size.

One may object that our practice of assigning a constant PIN for each quarter effectively reduces the time series variation of VPIN. To address this concern, we average the time series of VTPIN estimated from our extended framework to form quarterly series that is comparable with the quarterly VPIN series. By running the quarterly panel regressions in this way, we sacrifice some time series variation for VTPIN. The results are presented in Panel (B) of Table 6, and they are qualitatively the same as the daily results presented above. The only difference is that VPIN's interactions with δ_2 and δ_3 are significant only in the case of opening spread and remain positive. The results for VTPIN is still strong and robust. In sum, the evidence suggests that the time-varying probability of informed trading TPIN is a better and more powerful measure of information asymmetry than the constant PIN.

5.2 Predictive Power of Competing Measures of Information Asymmetry

Given that our extended model is computationally involved, we want to examine if the time-varying probability of informed trading is more informative when compared to other measures of information asymmetry. Therefore, we design the following fixed-effect panel regression as a horse race among competing measures of information asymmetry in predicting the mean spread for the next trading day,

$$\begin{aligned}\Sigma_{i,t+1} = & \beta_{0,i} + \beta_1 \cdot \text{VTPIN}_{i,t} + \beta_2 \cdot \text{VPIN}_{i,t} + \beta_3 \cdot \text{VOL}_{i,t} + \beta_4 \cdot \text{AMS}_{i,t} \\ & + \beta_5 \cdot \text{AVOL}_{i,t} + \beta_6 \cdot \text{OIMB}_{i,t} + \beta_7 \cdot \text{ME}_{i,t} + \beta_8 \cdot \text{RVOL}_{i,t} + \beta_9 \cdot \delta_2(t) \\ & + \beta_{10} \cdot \delta_3(t) + \beta_{11} \cdot \delta_2(t) \cdot \text{VTPIN}_{i,t} + \beta_{12} \cdot \delta_2(t) \cdot \text{VPIN}_{i,t} \\ & + \beta_{13} \cdot \delta_3(t) \cdot \text{VTPIN}_{i,t} + \beta_{14} \cdot \delta_3(t) \cdot \text{VPIN}_{i,t} + \eta_{i,t+1},\end{aligned}\tag{8}$$

where $i = 1, \dots, 40$. The dependent variable Σ is the mean bid-ask spread computed from the NYSE TAQ database. VTPIN, VPIN and VOL are the same as defined earlier. AMS is the abnormal mean spread computed as the deviation of current mean bid-ask spread from the moving average of past 20-trading-day mean bid-ask spread. AVOL is the abnormal volume computed as the deviation of current dollar volume from the moving average of past 20-trading-day mean dollar volume. OIMB is the order imbalance or absolute net order flow in number of trades. ME is the market value of equity. RVOL is the volatility of returns in the past 20-trading-day period. $\delta_2(t)$ and $\delta_3(t)$ are the tick size dummies as defined earlier.

For the same reasons we discussed in the previous sub-section, the dollar volume VOL and tick size dummies δ_2 and δ_3 are expected to have negative signs for their

respective coefficient. Each of the five competing measures of information asymmetry, including VTPIN, VPIN, AMS, AVOL and OIMB, should have a positive sign when predicting the future spread alone. Again we have no prior on how the pattern of signs will play out once all of them enter the same regression. Some of them may become insignificant if they do not contain additional information other than what the dominant measures already have. Or, they may appear significant jointly, perhaps because they may point to different dimensions of information asymmetry. Chordia, Roll, and Subrahmanyam (2002) find that order imbalances reduce liquidity, so the predicted sign for absolute order imbalance is positive. Large stocks tend to be more liquid so it is reasonable to conjecture a negative coefficient associated with market equity ME. Inventory theory suggests that a risk-averse market maker will set higher spread for stocks with higher past return volatility, so the expected sign for RVOL is positive.

The panel regression results in Panel (A) of Table 7 indicate that all the explanatory variables are highly significant and have the same sign as expected. The only exception is that the absolute order imbalance has a significant yet negative sign. The correlation between the absolute order imbalance and the contemporaneous mean spread is -0.21 , and the correlation between the absolute order imbalance and the next day mean spread is also -0.21 . The interaction terms between VTPIN and the dummies δ_2 and δ_3 are both significantly negative, with an intensified effect in the post-decimalization period. These results are consistent with the contemporaneous results and support the predictive power of the time-varying probability of informed trading TPIN. VPIN's interaction terms with δ_2 and δ_3 are both significantly positive, contrary to the empirical predictions. The magnitude of the two coefficients seem to be inconsistent internally, because a positive interaction with δ_2 should imply a strengthened positive interaction with δ_3 . That is, if the tick size reduction to $\frac{1}{16}$ (with $\delta_2 = 1$ and $\delta_3 = 0$) is expected to intensify the VPIN's impact on future mean spread, then a stronger tick size reduction to $\frac{1}{100}$ (with $\delta_2 = 0$ and $\delta_3 = 1$) should induce more impact of VPIN on future mean spread. Results in Panel (A) show a smaller impact (0.002) for the post-decimalization period than the impact (0.003) for the $\frac{1}{16}$ tick size period. This result casts some doubt on the consistency of VPIN in measuring information asymmetry.

We also perform the fixed-effect panel regressions by averaging daily series into quarterly ones. The results in Panel (B) are qualitatively similar to results in Panel (A). The VTPIN measure is again better than the VPIN measure in predicting spreads. We conclude that the time-varying probability of informed trading TPIN is a better and more robust measure in predicting future mean spread, even after controlling for other competing measures of information asymmetry.

5.3 Volume, Information Asymmetry and Autocorrelation of Returns

There are many reasons why investors trade. They may trade because they have information regarding the value of the stock, or they may trade for non-informational reasons. For example, they may trade to rebalance portfolios in order to better diversify risk. They also may trade because they are subject to behavioral biases such as overconfidence. Therefore, trading volume is often watched carefully by various traders. Not surprisingly, there is a large literature in finance devoted to volume (see, for example, Blume, Easley and O'Hara (1994), Lee and Swaminathan (2000), and Lo and Wang (2000)).

Campbell, Grossman and Wang (1993) investigated the relationship between stock market trading volume and serial correlation of daily stock returns. They found that the first-order daily return autocorrelation tends to decline with volume. They proposed a model with risk-averse "market makers" who charge a premium for accommodating "liquidity" or "non-informational" traders. The resulting changing expected returns reward market makers for playing this role. Therefore, the stock price decline on a high-volume day is more likely than the stock price decline on a low-volume day to be associated with an increase in the expected stock return.

More recently, Llorente et al. (2002) analyzed the dynamic relation between daily volume and first-order autocorrelation for individual stock returns. They presented a model in which returns generated by non-informational trades tend to reverse themselves, while returns generated by informational trades tend to continue themselves. This relationship is intuitive in that the stock prices reflect new information revealed by informed trades only in a gradual fashion. In the days immediately after the informed trades, stock prices tend to continue its decline (if the informed trades revealed bad news) or rise (if the informed trades revealed good news). On the other hand, large volumes of uninformed trades cause only short-lived price pressure so it is more likely to see a price reversal after uninformed trades. Their empirical tests show that the cross-sectional variation in the relation between volume and return autocorrelation is positively related to the extent of informed trading.

Specifically, Llorente et al. (2002) ran the following time-series regression at the first stage for each of the 2,226 stocks in their sample,

$$r_{i,t+1} = \gamma_0 + r_{i,t}(\gamma_1 + \gamma_2 \cdot \text{VOL}_{i,t}) + \varepsilon_{i,t+1}, \quad (9)$$

where the dependent variable r is the daily holding period return from CRSP, and the independent variable VOL is the detrended log turnover as a volume measure. They then ran a second-stage cross-sectional regression to use different proxies of

information asymmetry to explain the estimated γ_2 coefficient. A positive relation between the estimated γ_2 coefficient and measures of information asymmetry is interpreted as supporting evidence for the hypothesis that information asymmetry drives return autocorrelation higher.

We tried to replicate their results using our sample. However, the estimated γ_2 coefficient using their volume measure turned out mostly insignificant for the first stage of regressions. As is evident from their categorical analysis (in their Tables 2 and 3), a large fraction of stocks in their sample did not have significant γ_2 coefficient either. Our sample size of 40 stocks is too small to have enough power to replicate their results, so we decided to run a fixed-effect panel regression instead of the two-stage regression approach. The basic design is

$$r_{i,t+1} = \gamma_{0,i} + r_{i,t}(\gamma_1 + \gamma_2 \cdot ia_{i,t}) + \varepsilon_{i,t+1}, \text{ where } i = 1, \dots, 40. \quad (10)$$

We use three measures of information asymmetry (denoted by ia): dollar volume VOL, daily mean spread MSPR and VTPIN. The first two measures are proxies similar to what were used in Llorente et al. (2002), and we argue that VTPIN is a direct measure of information asymmetry due to evidence presented earlier. We predict a positive γ_2 since higher level of information asymmetry induces higher return autocorrelation.

Although volume is no longer an explanatory variable in the design above as it is in Llorente et al. (2002), we can still study the impact of volume on the return auto-correlation. This goal is achieved indirectly by comparing the γ_2 coefficient across different turnover deciles. It is plausible to conjecture that stocks in the high turnover decile are likely to have lower fraction of informed trading, so a second empirical prediction is that the γ_2 coefficient should decline as we move to higher turnover deciles. To test this specific hypothesis, we augment the empirical design with three dummies τ_4 , τ_6 and τ_8 , one for each of the 4th, 6th, and 8th turnover deciles, respectively, so that

$$r_{i,t+1} = \gamma_{0,i} + r_{i,t} \cdot [\gamma_1 + (\gamma_{2,2} + \gamma_{2,4} \cdot \tau_4 + \gamma_{2,6} \cdot \tau_6 + \gamma_{2,8} \cdot \tau_8) \cdot ia_{i,t}] + \varepsilon_{i,t+1}. \quad (11)$$

The regression results in Table 8 show a statistically significant negative autocorrelation γ_1 for daily stock returns. Ignoring the difference in turnover deciles, the γ_2 coefficient is positive but insignificant when either VOL or MSPR is used. This is in sharp contrast to the significantly positive γ_2 associated with VTPIN. After introducing the turnover group dummies, the γ_2 coefficients are not significant for any turnover group when VOL is used, and the γ_2 coefficients are significant for all four turnover group when MSPR is used. In the latter case, the γ_2 for the

second turnover group is negative, and it is increasingly positive as we move to higher turnover decile. So neither VOL nor MSPR conforms to the empirical predictions. When VTPIN is used, the γ_2 coefficients are highly significant and positive for the 2nd, 6th, and 8th turnover deciles. We also observe the monotonically declining impact of information asymmetry on return auto-correlation as the turnover decile becomes higher, as the γ_2 coefficients are 0.0172, 0.0127, 0.0103 and 0.0099 for the 2nd, 4th, 6th and 8th turnover decile, respectively. In sum, the time series of VTPIN enables us to conduct a sharper test, providing strong support for the argument of Llorente et al. (2002).

The time series of probability of informed trading also can be used in studies investigating if the information asymmetry risk is priced and if it affects asset returns. For example, Easley, Hvidkjaer, and O'Hara (2002) repeatedly estimated the probability of informed trading for each year in their sample period using the same framework as Easley et al. (1996) and reach the conclusion that a 10% difference in this probability between two stocks leads to a 2.5% difference in annual returns. Their analysis could be extended with the time-varying probability of informed trading. We leave this for future research.

6 Concluding Remarks

Building upon the seminal work of Easley et al. (1996), we develop a framework to investigate time-varying informed and uninformed trading activities and the relationship between them. We allow the buy and sell arrival rates for the uninformed traders to follow a Markov switching process. Both the uninformed buy-sell arrival rate differential and the uninformed transition probabilities depend on past performance of the stock and the overall market. The informed traders may match the level of the uninformed arrival rate with certain probability so as to make better use of the camouflage provided by the uninformed traders.

Our empirical estimation of 40 NYSE stocks shows that the buy and sell arrival rates of the uninformed traders are different and time-varying. The uninformed traders tend to adopt contrarian strategy in reaction to high prior own stock returns, but employ momentum strategy in reaction to high prior market returns. Informed traders seem to take good advantage of the camouflage since the estimated probability of informed matching response ranges from 0.72 to 0.98. We find that the estimated time-varying probability of informed trading is a good predictor for various measures of bid-ask spreads, and is a better and more powerful measure of information asymmetry than the constant probability of informed trading. The estimated time-varying probability of informed trading has predictive power over mean bid-ask

spreads in the next trading day, even after controlling for competing measures of information asymmetry. Finally, we use the estimated time-varying probability of informed trading as a measure of informational asymmetry to analyze its impact on the serial correlation of daily stock returns. The development of this time series enables us to conduct a sharper and more robust test on this important issue.

7 Appendix

7.1 Probability of Informed Trading

Using similar arguments in deriving equation (4), we can obtain other elements of the transition matrix for the four-composite-state model. The matrix ν_t of transition probabilities can be written as follows,

$$\nu_t \equiv \begin{bmatrix} \pi_t^{**}\rho & \pi_t^{**}(1-\rho) & (1-\pi_t^{**})(1-\rho) & (1-\pi_t^{**})\rho \\ \pi_t^{**}\rho & \pi_t^{**}(1-\rho) & (1-\pi_t^{**})(1-\rho) & (1-\pi_t^{**})\rho \\ (1-\pi_t^*)\rho & (1-\pi_t^*)(1-\rho) & \pi_t^*(1-\rho) & \pi_t^*\rho \\ (1-\pi_t^*)\rho & (1-\pi_t^*)(1-\rho) & \pi_t^*(1-\rho) & \pi_t^*\rho \end{bmatrix}, \quad (12)$$

where each row stands for a composite state at trading day $t-1$ and each column stands for a composite state at trading day t . Note that the first and second rows in this transition matrix are the same, as are the third and fourth rows. This result comes from the assumption that the probability of matching response is path-independent. In particular, the informed traders form their response ρ not based upon the state level of the uninformed arrival rate in the previous trading day, and the uninformed transition probabilities are the same across these rows.

Denote as $\mathbf{p}_t^s$ the vector of probabilities for each composite state at period t

$$\mathbf{p}_t^s \equiv [p(s_t^a) \ p(s_t^b) \ p(s_t^c) \ p(s_t^d)].$$

The evolution of these state probabilities can be written as

$$\mathbf{p}_t^s = \mathbf{p}_{t-1}^s \nu_t = \mathbf{p}_0^s \prod_{m=1}^t \nu_m, \quad \forall t \geq 1, \quad (13)$$

where we assume $\mathbf{p}_0^s = [0.25 \ 0.25 \ 0.25 \ 0.25]$, without loss of generality.

By combining the elements of $\mathbf{p}_t^s$, we can obtain the probabilities $p(\varepsilon_t^h)$, $p(\varepsilon_t^l)$, $p(\mu_t^h)$ and $p(\mu_t^l)$. We then define the expectation of the uninformed and the informed arrival rates as

$$\varepsilon_t^{e,B} \equiv \varepsilon_t^{h,B} \cdot p(\varepsilon_t^h) + \varepsilon_t^{l,B} \cdot p(\varepsilon_t^l), \quad \varepsilon_t^{e,S} \equiv \varepsilon_t^{h,S} \cdot p(\varepsilon_t^h) + \varepsilon_t^{l,S} \cdot p(\varepsilon_t^l), \quad \text{and} \quad \mu_t^e \equiv \mu_t^h \cdot p(\mu_t^h) + \mu_t^l \cdot p(\mu_t^l).$$

Finally, the probability that each trade on day t is information-based can be expressed as

$$\text{TPIN}_t = \frac{\alpha \mu_t^e}{\varepsilon_t^{e,B} + \varepsilon_t^{e,S} + \alpha \mu_t^e}. \quad (14)$$

7.2 Sample Likelihood

Conditional on the state of a trading day t , say s_t^a , the likelihood of observing B_t buy trades and S_t sell trades is

$$\begin{aligned} g(s_t^a) = & (1 - \alpha) \exp \left(-\varepsilon_t^{h,B} - \varepsilon_t^{h,S} \right) \frac{\left(\varepsilon_t^{h,B} \right)^{B_t}}{B_t!} \frac{\left(\varepsilon_t^{h,S} \right)^{S_t}}{S_t!} \\ & + \alpha \delta \exp \left(-\varepsilon_t^{h,B} - \varepsilon_t^{h,S} - \mu_t^h \right) \frac{\left(\varepsilon_t^{h,B} \right)^{B_t}}{B_t!} \frac{\left(\varepsilon_t^{h,S} + \mu_t^h \right)^{S_t}}{S_t!} \\ & + \alpha (1 - \delta) \exp \left(-\varepsilon_t^{h,B} - \varepsilon_t^{h,S} - \mu_t^h \right) \frac{\left(\varepsilon_t^{h,B} + \mu_t^h \right)^{B_t}}{B_t!} \frac{\left(\varepsilon_t^{h,S} \right)^{S_t}}{S_t!}. \end{aligned} \quad (15)$$

As the number of trades gets larger, this conditional likelihood becomes harder to compute due to the factorial, the exponential and the power functions. To improve computational efficiency, we rewrite it as follows,

$$g(s_t^a) = c_t \cdot m(s_t^a) \cdot h(s_t^a) / (B_t! S_t!), \quad (16)$$

where the common factor c_t makes $\ln(c_t)$ easy to compute, and the state-dependent factors $m(s_t^a)$ and $h(s_t^a)$ are constructed to moderate the size of the exponential functions and the power functions. The definition of these factors are:

$$\begin{aligned} c_t & \equiv \exp \left(-\varepsilon_t^{*,B} - \varepsilon_t^{*,S} - \mu_t^* \right) \left(\varepsilon_t^{*,B} \right)^{B_t} \left(\varepsilon_t^{*,S} \right)^{S_t}, \\ m(s_t^a) & \equiv \exp \left[-(\varepsilon_t^{h,B} - \varepsilon_t^{*,B}) - (\varepsilon_t^{h,S} - \varepsilon_t^{*,S}) - (\mu_t^h - \mu_t^*) \right] \left(\frac{\varepsilon_t^{h,B}}{\varepsilon_t^{*,B}} \right)^{B_t} \left(\frac{\varepsilon_t^{h,S}}{\varepsilon_t^{*,S}} \right)^{S_t}, \\ h(s_t^a) & \equiv \left[(1 - \alpha) \exp(\mu_t^h) + \alpha \delta \left(1 + \frac{\mu_t^h}{\varepsilon_t^{h,S}} \right)^{S_t} + \alpha (1 - \delta) \left(1 + \frac{\mu_t^h}{\varepsilon_t^{h,B}} \right)^{B_t} \right], \\ \varepsilon_t^{*,B} & \equiv \frac{1}{2}(\varepsilon_t^{h,B} + \varepsilon_t^{l,B}), \\ \varepsilon_t^{*,S} & \equiv \frac{1}{2}(\varepsilon_t^{h,S} + \varepsilon_t^{l,S}), \\ \mu_t^* & \equiv \frac{1}{2}(\mu_t^h + \mu_t^l). \end{aligned} \quad (17)$$

Using a similar method, we also can rewrite and calculate the conditional likelihood for other state types, namely $g(s_t^b)$, $g(s_t^c)$ and $g(s_t^d)$. Taking into account the probabilities for each state type, the unconditional likelihood for day t is

$$L\{(B_t, S_t)|\Theta\} = \sum_{k \in \{a,b,c,d\}} g(s_t^k) \cdot p(s_t^k), \quad (18)$$

where Θ is the vector of parameters to be estimated. By purging some constants that do not contain parameters to be estimated, we can write the sample log-likelihood function over a sample of T trading days in the following computation-friendly form,

$$\begin{aligned} \mathcal{L}(\Theta) &= \ln \left[\prod_{t=1}^T L\{\Theta|B_t, S_t\} \right] \\ &= \sum_{t=1}^T \left[-\varepsilon_t^{*,B} - \varepsilon_t^{*,S} - \mu_t^* + B_t \ln \left(\varepsilon_t^{*,B} \right) + S_t \ln \left(\varepsilon_t^{*,S} \right) \right] \\ &\quad + \sum_{t=1}^T \ln \left[\sum_{k \in \{a,b,c,d\}} m(s_t^k) h(s_t^k) p(s_t^k) \right]. \end{aligned} \quad (19)$$

References

- [1] Admati, Anat R., 1991, The Informational Role of Prices, *Journal of Monetary Economics* 28, 347-361.
- [2] Admati, Anat R., and Paul Pfleiderer, 1988, A Theory of Intraday Patterns: Volume and Price Variability, *Review of Financial Studies* 1, 3-40.
- [3] Angel, James J., 1997, Tick Size, Share Prices, and Stock Splits, *Journal of Finance* 52, 655-681.
- [4] Back, Kerry, 1992, Insider Trading in Continuous Time, *Review of Financial Studies* 5, 387-409.
- [5] Back, Kerry, and Hal Pedersen, 1998, Long-lived Information and Intraday Patterns, *Journal of Financial Markets* 1, 385-402.
- [6] Bagehot, Walter, 1971, The Only Game in Town, *Financial Analysts Journal* 27, 12-14, 22.
- [7] Barber, Brad M. and Terrance Odean, 2003, All that Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors, *University of California at Davis Working Paper*.
- [8] Barberis, Nick, and Ming Huang, 2001, Mental Accounting, Loss Aversion, and Individual Stock Returns, *Journal of Finance* 56, 1247-1292.

- [9] Barberis, Nick, Ming Huang, and Tano Santos, 2001, Prospect Theory and Asset Prices, *Quarterly Journal of Economics* 116, 1-53.
- [10] Barberis, Nicholas, Andrei Shleifer, and Robert Vishny, 1998, A Model of Investor Sentiment, *Journal of Financial Economics* 49, 307-343.
- [11] Barclay, Michael J., Robert H. Litzenberger, and Jerold B. Warner, 1990, Private Information, Trading Volume, and Stock-return Variances, *Review of Financial Studies* 3, 233-253.
- [12] Blume, Lawrence, David Easley, and Maureen O'hara, 1994, Market Statistics and Technical Analysis - the Role of Volume, *Journal of Finance* 49, 153-181.
- [13] Caginalp, Gunduz, David Porter, and Vernon Smith, 2000, Momentum and Overreaction in Experimental Asset Markets, *International Journal of Industrial Organization* 18, 187-204.
- [14] Campbell, John Y, Sanford J. Grossman, and Jiang Wang, 1993, Trading Volume and Serial Correlation in Stock Returns, *Quarterly Journal of Economics*, 108, 905-939.
- [15] Chan, Louis K. C., Narasimhan Jegadeesh, and Josef Lakonishok, 1996, Momentum Strategies, *Journal of Finance* 51, 1681-1731.
- [16] Choe, Hyuk, Bong-Chan Kho and Rene M. Stulz, 1999, Do Foreign Investors Destabilize Stock Markets? The Korean Experience in 1997, *Journal of Financial Economics* 54, 227-264.
- [17] Chordia, Tarun, Richard Roll, and Avanidhar Subrahmanyam, 2002, Order Imbalance, Liquidity, and Market Returns, *Journal of Financial Economics* 65, 111-130.
- [18] Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam, 1998, Investor Psychology and Security Market Under- and Overreactions, *Journal of Finance* 53, 1839-1885.
- [19] Diamond, Douglas W., and Robert E. Verrecchia, 1981, Information Aggregation in a Noisy Rational Expectations Economy, *Journal of Financial Economics* 9, 221-235.
- [20] Easley, David, Robert F. Engle, Maureen O'Hara, and Liuren Wu, 2002, Time-varying Arrival Rates of Informed and Uninformed Trades, *Cornell University Working Paper*.
- [21] Easley, David, Soeren Hvidkjaer, and Maureen O'Hara, 2002, Is Information Risk a Determinant of Asset Returns?, *Journal of Finance* 57, 2185-2221.

- [22] Easley, David, Nicholas M. Kiefer, and Maureen O'Hara, 1997, One Day in the Life of a Very Common Stock, *Review of Financial Studies* 10, 805-835.
- [23] Easley, David, Nicholas M. Kiefer, Maureen O'Hara, and Joseph B. Paperman, 1996, Liquidity, Information, and Infrequently Traded Stocks, *Journal of Finance* 51, 1405-1436.
- [24] Easley, David, and Maureen O'Hara, 1987, Price, Trade Size, and Information in Securities Markets, *Journal of Financial Economics* 19, 69-90.
- [25] Easley, David, and Maureen O'Hara, 1992, Time and the Process of Security Price Adjustment, *Journal of Finance* 47, 577-605.
- [26] Garman, Mark B., 1976, Market Microstructure, *Journal of Financial Economics* 3, 257-275.
- [27] Glosten, Lawrence R., and Lawrence E. Harris, 1988, Estimating the Components of the Bid/Ask Spread, *Journal of Financial Economics* 21, 123-42.
- [28] Glosten, Lawrence R., and Paul R. Milgrom, 1985, Bid, Ask, and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders, *Journal of Financial Economics* 14, 71-100.
- [29] Griffin, John M., Jeffrey H. Harris, and Selim Topaloglu, 2003, The Dynamics of Institutional and Individual Trading, *Journal of Finance* 58, 2285-2320.
- [30] Grinblatt, Mark and Matti Keloharju, 2001, What Makes Investors Trade?, *Journal of Finance* 56, 589-616.
- [31] Hasbrouck, Joel, 1991a, Measuring the Information Content of Stock Trades, *Journal of Finance* 46, 179-207.
- [32] Hasbrouck, Joel, 1991b, The Summary Informativeness of Stock Trades: An Econometric Analysis, *Review of Financial Studies* 4, 571-595.
- [33] Hong, Harrison, and Jeremy C. Stein, 1999, A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets, *Journal of Finance* 54, 2143-2184.
- [34] Jegadeesh, Narasimhan, 1990, Evidence of Predictable Behavior of Security Returns, *Journal of Finance* 45, 881-898.
- [35] Jegadeesh, Narasimhan, and Sheridan Titman, 1993, Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency, *Journal of Finance* 48, 65-91.
- [36] Jegadeesh, Narasimhan, and Sheridan Titman, 1995, Overreaction, Delayed Reaction, and Contrarian Profits, *Review of Financial Studies* 8, 973-993.

- [37] Jegadeesh, Narasimhan, and Sheridan Titman, 2001, Profitability of Momentum Strategies: An Evaluation of Alternative Explanations, *Journal of Finance* 56, 699-720.
- [38] Jones, Charles M., Gautam Kaul, and Marc L. Lipson, 1994, Information, Trading, and Volatility, *Journal of Financial Economics* 36, 127-154.
- [39] Kyle, Albert S., 1985, Continuous Auctions and Insider Trading, *Econometrica* 53, 1315-1335.
- [40] Lakonishok, Josef, Andrei Shleifer, and Robert W. Vishny, 1994, Contrarian Investment, Extrapolation, and Risk, *Journal of Finance* 49, 1541-1578.
- [41] Lee, Charles M. C., and Mark J. Ready, 1991, Inferring Trade Direction from Intraday Data, *Journal of Finance* 46, 733-746.
- [42] Lee, Charles M. C., and Bhaskaran Swaminathan, 2000, Price Momentum and Trading Volume, *Journal of Finance* 55, 2017-2069.
- [43] Lehmann, Bruce N., 1990, Fads, Martingales, and Market Efficiency, *Quarterly Journal of Economics* 105, 1-28.
- [44] Llorente, Guillermo, Roni Michaely, Gideon Saar, and Jiang Wang, 2002, Dynamic Volume-Return Relation of Individual Stocks, *Review of Financial Studies*, 15, 1005-1047.
- [45] Lo, Andrew W., and A. Craig MacKinlay, 1990, When Are Contrarian Profits Due to Stock Market Overreaction?, *Review of Financial Studies* 3, 175-205.
- [46] Lo, Andrew W., and Jiang Wang, 2000, Trading Volume: Definitions, Data Analysis, and Implications of Portfolio Theory, *Review of Financial Studies* 13, 257-300.
- [47] Madhavan, Ananth, and Seymour Smidt, 1991, A Bayesian Model of Intraday Specialist Pricing, *Journal of Financial Economics* 30, 99-134.
- [48] Madhavan, Ananth, and Seymour Smidt, 1993, An Analysis of Changes in Specialist Inventories and Quotations, *Journal of Finance* 48, 1595-1628.
- [49] Madhavan, Ananth, and George Sofianos, 1998, An Empirical Analysis of NYSE Specialist Trading, *Journal of Financial Economics* 48, 189-210.
- [50] Mendelson, Haim, 1982, Market Behavior in a Clearing House, *Econometrica* 50, 1505-1524.
- [51] Odean, Terrance, 1998, Are Investors Reluctant to Realize Their Losses, *Journal of Finance* 53, 1775-1798.

- [52] Odean, Terrance, 1999, Do Investors Trade Too Much? *American Economic Review* 89, 1279-1298.
- [53] O'Hara, Maureen, and George S. Oldfield, 1986, The Microeconomics of Market Making, *Journal of Financial and Quantitative Analysis* 21, 361-376.
- [54] Rouwenhorst, K. Geert, 1999, International Momentum Strategies, *Journal of Finance* 53, 267-284.
- [55] Siegel, Jeremy J., 1992, Equity Risk Premia, Corporate Profit Forecasts, and Investor Sentiment around the Stock Crash of October 1987, *Journal of Business* 65, 557-570.
- [56] Smidt, Seymour, 1971, Which Road to an Efficient Stock Market: Free Competition or Regulated Monopoly?, *Financial Analyst Journal* 27, 18-20, 64-69.
- [57] Wang, Jiang, 1993, A Model of Intertemporal Asset Prices under Asymmetric Information, *Review of Economic Studies* 60, 249-282.
- [58] Weston, James P., 2001, Information, Liquidity, and Noise, *Rice University Working Paper*.
- [59] Zabel, Edward, 1981, Competitive Price Adjustment without Market Clearing, *Econometrica* 49, 1201-1221.

Table 1. Summary Statistics

In this table, turnover is the ratio of daily share volume to shares outstanding (in basis points); prc is the average daily dollar price; nbuy is the average number of daily buy orders; nsell is the average number of daily sell orders; and cdr is the cumulative prior-20-trading-day stock return net of 3-month T-bill rate (in basis points). The sample period is from January 1, 1993 to December 31, 2002.

	Ticker	Company Name	Mean					Standard Deviation				
			turnover	prc	nbuy	nsell	cdr	turnover	prc	nbuy	nsell	cdr
2 nd Turnover Decile	EBF	Ennis Business Forms Inc	18.51	11.16	9.85	9.72	13.11	23.94	2.51	13.10	10.77	645.05
	GLO	G L Industries Inc	14.83	8.45	3.32	3.73	88.92	22.20	3.49	4.44	4.13	1209.73
	LYDL	Lydall Inc	18.60	18.90	9.20	9.73	32.58	27.83	8.32	9.79	9.31	1062.90
	OCQ	Oneida Ltd	15.46	17.89	9.28	9.91	71.45	20.45	5.83	9.85	8.12	1083.47
	OXM	Oxford Manufacturing Inc	17.53	23.31	7.55	7.64	45.60	21.00	5.50	7.14	6.69	933.57
	RUS	Russ Berrie & Co	16.02	21.28	12.92	13.51	91.24	20.90	6.29	13.58	11.90	888.61
	SFR	Standard Financial Corp	15.18	17.03	7.87	7.81	137.97	17.94	7.93	10.56	9.22	754.79
	SMP	Standard Motor Products Inc	16.16	17.00	8.23	8.10	37.91	20.02	4.63	7.75	6.55	1035.40
	TYL	Tyler Corp De	16.17	4.19	9.35	8.81	92.60	29.30	1.93	14.66	10.59	1758.24
	VCD	Value City Department Stores Inc	15.21	10.28	11.29	10.21	-66.53	20.92	4.48	10.88	8.46	1779.90
		GROUP AVERAGE	16.36	14.95	8.89	8.92	54.48	22.45	5.09	10.17	8.57	1115.17
4 th Turnover Decile	AGL	Angelica Corp	24.83	18.70	11.62	10.73	33.82	37.19	6.08	15.34	12.29	871.55
	BW	Brush Wellman Inc	24.26	16.96	14.21	14.16	-56.52	40.07	4.28	15.74	13.76	994.01
	CAFE	Cafeterias Inc	25.89	16.17	19.46	19.90	-109.53	26.23	6.69	12.12	11.70	890.67
	CPIC	C P I Corp	25.35	19.83	9.50	9.88	9.49	34.57	4.46	8.92	7.22	1017.92
	DMAL	Dayton Malleable Inc	25.73	16.36	7.60	7.74	-110.20	29.58	6.41	6.83	6.36	1178.78
	DVIC	Diagnostic Ventures Inc	25.07	13.97	10.61	10.80	52.12	40.34	4.03	13.74	11.42	1356.70
	IMC	International Multifoods Corp	25.80	20.97	19.41	18.70	30.19	27.14	4.24	20.56	17.90	924.04
	PAWN	Cash America Investments Inc	27.08	9.06	17.12	17.14	28.98	35.36	2.70	17.73	13.93	1198.64
	RML	Russell Mls Inc	24.71	24.27	31.05	29.35	-24.23	33.37	6.40	24.51	18.34	904.57
	UNFI	Unifi Inc	25.13	21.44	32.43	30.13	-93.00	29.70	10.66	27.70	22.87	1172.08
		GROUP AVERAGE	25.39	17.77	17.30	16.85	-23.89	33.35	5.59	16.32	13.58	1050.90
0 th Turnover Decile	AIR	A A R Corp	32.15	18.03	22.51	20.83	52.65	37.58	7.98	21.51	16.62	1274.53
	BDG	Bandag Inc	32.86	43.81	23.60	20.28	-26.44	31.45	11.72	21.42	15.76	810.45
	ESL	Esterline Corp	33.77	18.90	21.36	19.63	117.07	40.39	8.05	25.38	21.70	1037.68
	FRBC	First Republic Bancorp Inc	34.47	20.79	11.85	11.13	105.93	55.73	7.23	16.34	14.48	981.31

8 th Turnover Decile	HEXC	Hexcel Corp	36.51	10.50	15.38	15.01	66.82	60.26	6.93	15.47	13.75	1909.97
	HUF	Huffman Mfg Co	32.93	12.22	9.65	11.07	5.03	47.18	4.12	11.06	8.66	1311.69
	LZ	Lubrizol Corp	33.85	31.12	58.03	54.65	27.58	28.92	4.86	48.11	42.31	766.79
	NCO	North American Coal Corp	33.96	63.82	24.46	22.09	38.93	44.95	24.18	21.30	18.98	1125.52
	OBOD	Owens Minor & Bodeker Inc	31.60	15.04	27.66	23.24	73.23	37.54	3.86	30.80	21.66	1018.44
	RTI	R M I Titanium Co	34.65	12.18	19.99	19.09	149.16	56.52	7.32	20.64	17.41	1659.39
	GROUP AVERAGE		33.67	24.64	23.45	21.70	61.00	44.05	8.63	23.20	19.13	1189.58
	ALN	Allen Elec & Equip Co	45.03	17.95	29.63	26.37	88.82	53.76	9.73	28.07	18.89	1688.84
	BGG	Briggs & Stratton Corp	44.26	49.18	54.83	47.05	71.67	41.55	14.18	43.68	34.41	849.89
	GSR	Green Shoe Mfg Co	46.40	10.61	39.27	37.71	-15.12	46.85	3.58	28.94	20.45	1202.10
	KMT	Kennametal Inc	44.73	34.68	48.75	43.27	119.46	53.01	8.82	46.30	39.14	1087.42
	NEV	Nuevo Energy Co	49.74	23.20	24.98	23.40	39.59	65.86	10.35	23.83	19.58	1445.05
	OCER	Oceaneering International Inc	44.42	16.56	35.07	31.91	110.66	40.42	4.74	44.18	38.21	1301.12
	OS	Oregon Steel Mills Inc	41.23	13.76	22.83	22.95	13.56	50.61	7.18	19.45	15.60	1877.30
	POP	Pope & Talbot Inc	39.81	16.19	18.22	16.17	48.62	48.11	4.65	19.35	14.01	1131.10
	WLMN	Wellman Inc	41.18	19.08	37.25	34.68	29.87	36.85	5.13	30.13	22.16	1146.50
	WLV	Wolverine Shoe & Tanning	45.94	20.71	40.31	35.15	174.07	49.37	7.82	39.19	31.70	1134.33
	GROUP AVERAGE		44.27	22.19	35.11	31.87	68.12	48.64	7.62	32.31	25.42	1286.36

Table 2. Maximum Likelihood Estimation Results

In this table, α is the probability of news arrival; δ is the probability of the arrived news being negative; $\varepsilon^{h,B}$ is the uninformed arrival rates for buyers at the high state; $\varepsilon^{l,B}$ is the uninformed arrival rates for buyers at the low state; μ^h is the informed arrival rate at the high state; μ^l is the informed arrival rate at the low state; ρ is the probability of informed matching response; N is the number of daily observations used; and $\mathcal{L}/N$ is the mean log-likelihood net of a stock-specific constant. All the estimated parameters in this table are statistically significant at the 1% level.

	Ticker	α	δ	$\varepsilon^{h,B}$	$\varepsilon^{l,B}$	μ^h	μ^l	ρ	N	$\mathcal{L}/N$
2 nd Turnover Decile	EBF	0.3606	0.3407	25.3068	4.6134	35.6220	7.3950	0.9774	2494	31.2030
	LDL	0.4325	0.4652	18.3684	4.2307	24.1162	7.2877	0.9342	2489	27.8817
	OCQ	0.3668	0.2810	13.5830	2.9912	21.7897	6.9311	0.8521	2496	28.3652
	OXM	0.3597	0.4475	14.5257	3.8086	18.2192	6.6033	0.9337	2489	18.0875
	RUS	0.3860	0.3086	25.3295	5.3118	33.0770	9.8350	0.9415	2495	48.0752
	SL	0.3056	0.4641	8.8995	1.6742	15.5959	4.0547	0.9698	2376	3.3616
	SMP	0.3697	0.3453	14.1699	3.8184	19.5877	6.8686	0.9463	2488	20.4594
	STL	0.3232	0.3605	18.6515	2.5730	24.9547	6.6841	0.9519	2453	22.5024
	TYL	0.3113	0.3059	19.9556	2.3166	41.3703	8.0879	0.9294	2469	29.5764
	VCD	0.4278	0.3100	19.3811	4.6236	24.1846	8.0623	0.8726	2496	33.4767
4 th Turnover Decile	AGL	0.2846	0.3967	39.6797	6.7863	55.9571	10.3353	0.9826	2495	38.2509
	AIZ	0.4472	0.2782	11.6960	3.4960	18.3268	6.2869	0.8881	2494	17.9581
	BW	0.3679	0.3039	29.6577	5.9326	40.6657	10.4433	0.9547	2496	54.1906
	CPY	0.3877	0.4638	19.3965	5.4935	28.0686	7.9290	0.9528	2494	27.1531
	DVI	0.3513	0.3785	25.5340	4.3949	40.9151	9.6319	0.9431	2491	35.2363
	IMC	0.3779	0.3060	41.8606	8.5806	46.7272	13.1042	0.9167	2495	84.2299
	LUB	0.5151	0.2453	26.6487	11.1107	30.7906	10.7440	0.8697	2496	81.0577
	PWN	0.4156	0.3281	31.3961	7.8103	44.8537	11.6299	0.9007	2496	70.3109
	RML	0.5128	0.2336	46.5156	13.7252	50.5028	17.2250	0.8594	2496	153.9394
	UFI	0.4886	0.2503	49.4225	13.8094	62.1200	19.0889	0.8327	2496	164.2640
6 th Turnover Decile	AIR	0.4200	0.1784	31.1139	6.2559	45.4627	13.7534	0.8060	2496	99.6570
	BDG	0.4317	0.1739	40.9406	10.1428	48.0089	13.8305	0.8983	2496	99.9757
	ESL	0.4569	0.1613	44.0452	5.3286	47.7488	12.9937	0.8960	2495	98.2780
	FRC	0.3524	0.3164	29.2591	3.5008	36.4624	9.4723	0.9406	2472	43.1378
	HUF	0.4177	0.3333	22.2841	4.8808	42.2031	8.9773	0.9597	2496	31.2681
	HXL	0.4522	0.2673	23.6084	4.8736	32.0509	11.0737	0.8254	2490	59.3771
	LZ	0.4829	0.1497	96.2324	22.1353	72.4437	26.9893	0.8230	2496	365.8356
	NC	0.3994	0.3064	40.4336	9.2145	43.3457	14.3942	0.8991	2495	110.2241
	OMI	0.3998	0.2389	61.4338	10.3135	52.9010	14.6037	0.8999	2496	129.7873
	RTI	0.3652	0.3353	24.1150	3.3852	57.6502	14.7773	0.5860	2468	88.3287
8 th Turnover Decile	ALN	0.3952	0.3132	46.3659	13.1012	85.3335	18.7486	0.8276	2496	141.1333
	BGG	0.4032	0.2701	83.6650	25.5594	87.9183	27.0532	0.7922	2496	317.9753
	KMT	0.4500	0.2137	83.2251	17.3548	91.5251	26.7544	0.8684	2496	285.4606
	NEV	0.4155	0.2283	31.3696	5.2529	54.2020	14.8623	0.7187	2493	117.9120
	OII	0.4810	0.1367	100.5485	10.1795	65.8424	20.9984	0.8520	2496	195.2407
	OS	0.3907	0.4163	34.5097	11.3866	51.3294	15.4659	0.8276	2496	104.2254
	POP	0.4022	0.2378	34.9956	8.1640	48.6301	12.6672	0.9257	2496	71.3983
	SRR	0.4936	0.2814	51.8576	20.5507	69.8886	20.5316	0.8289	2496	212.9184
	WLMN	0.4260	0.2945	60.6199	20.2977	77.1817	21.2942	0.8935	2496	196.1643
	WWW	0.4290	0.3849	70.5638	14.0083	64.5101	21.7332	0.8196	2496	220.6804

Table 3. Estimated Mean Information Variables

In this table, $\bar{\varepsilon}_t^{e,B}$ is the sample mean expected uninformed arrival rate for buyers; $\bar{\varepsilon}_t^{e,S}$ is the sample mean expected uninformed arrival rate for sellers; $\bar{\mu}_t^e$ is the sample mean expected informed arrival rate; and $\overline{\text{TPIN}}_t^e$ is the sample mean expected probability of information-based trading. The numbers inside parentheses are the respective sample standard deviations.

	Ticker	$\bar{\varepsilon}_t^{e,B}$		$\bar{\varepsilon}_t^{e,S}$		$\bar{\mu}_t^e$		$\overline{\text{TPIN}}_t^e$	
2 nd Turnover Decile	EBF	7.5873	(4.4719)	8.9233	(5.0771)	11.9061	(5.8242)	0.2108	(0.0117)
	LDL	6.6714	(1.3062)	7.7066	(1.4302)	10.9181	(1.3501)	0.2485	(0.0141)
	OCQ	6.0672	(1.2155)	8.7891	(2.2865)	12.1676	(1.2006)	0.2346	(0.0194)
	OXM	5.8037	(0.7200)	6.2386	(0.9691)	9.2494	(0.6768)	0.2175	(0.0092)
	RUS	9.0671	(4.2019)	11.8110	(4.7386)	15.0445	(4.3081)	0.2245	(0.0189)
	SL	2.4011	(0.4880)	3.0531	(1.0677)	5.4945	(0.7323)	0.2390	(0.0161)
	SMP	5.9994	(0.1421)	6.8936	(0.5360)	9.9437	(0.1559)	0.2221	(0.0063)
	STL	5.6304	(1.4708)	6.5340	(1.7330)	10.7032	(1.5105)	0.2247	(0.0154)
	TYL	6.2354	(1.7551)	7.6360	(2.2881)	16.7878	(2.8441)	0.2779	(0.0194)
	VCD	7.8569	(2.4260)	8.7848	(2.1564)	12.7483	(1.9752)	0.2518	(0.0209)
4 th Turnover Decile	AGL	9.3329	(4.8364)	9.7831	(5.9505)	14.5366	(6.4750)	0.1823	(0.0105)
	AIZ	4.8490	(0.6286)	6.7003	(0.8245)	9.1760	(0.7165)	0.2631	(0.0089)
	BW	10.5383	(5.3167)	12.6098	(5.3752)	17.1476	(6.1592)	0.2196	(0.0133)
	CPY	7.3065	(0.7714)	8.2000	(0.9347)	11.2587	(1.0118)	0.2199	(0.0060)
	DVI	7.5695	(2.0637)	8.9868	(2.7018)	15.5747	(2.7066)	0.2509	(0.0147)
	IMC	14.2688	(0.4954)	16.3357	(2.5348)	20.6954	(0.4171)	0.2045	(0.0123)
	LUB	13.9198	(1.8489)	18.0392	(2.4724)	16.0364	(1.7636)	0.2058	(0.0066)
	PWN	12.3079	(5.0022)	15.2530	(5.2746)	20.0073	(5.6461)	0.2357	(0.0128)
	RML	21.3151	(0.7360)	26.6945	(4.0185)	27.4403	(0.5370)	0.2277	(0.0134)
	UFI	21.9633	(1.1647)	27.0927	(4.6087)	32.8437	(0.9364)	0.2474	(0.0171)
6 th Turnover Decile	AIR	14.7751	(3.8689)	18.9955	(3.9962)	26.5555	(3.0207)	0.2516	(0.0177)
	BDG	16.5084	(4.3995)	18.7208	(3.4331)	22.9341	(3.8892)	0.2209	(0.0087)
	ESL	14.5063	(10.9843)	18.7040	(11.0336)	23.1335	(7.8092)	0.2680	(0.0495)
	FRC	8.2224	(2.3797)	9.4833	(4.3251)	15.4352	(2.1973)	0.2403	(0.0197)
	HUF	6.3900	(1.0200)	9.6893	(2.7003)	12.9648	(1.7904)	0.2535	(0.0135)
	HXL	9.7972	(2.0324)	13.1517	(2.7297)	18.3239	(1.4811)	0.2681	(0.0191)
	LZ	43.2019	(21.6086)	53.0727	(21.8702)	43.3837	(8.5624)	0.1920	(0.0346)
	NC	18.5976	(3.0605)	19.2900	(2.5098)	24.2613	(2.2653)	0.2046	(0.0077)
	OMI	20.7801	(13.8308)	21.4695	(10.8513)	24.7095	(8.2863)	0.2040	(0.0283)
	RTI	14.3532	(0.2699)	16.7358	(2.6856)	36.4283	(0.0960)	0.3007	(0.0167)
8 th Turnover Decile	ALN	22.0793	(2.4836)	23.4850	(2.1264)	42.0026	(3.2572)	0.2673	(0.0069)
	BGG	42.3508	(0.8457)	43.6122	(7.4438)	49.9789	(0.5177)	0.1909	(0.0125)
	KMT	34.6692	(6.7084)	39.6791	(7.7414)	47.8212	(4.8608)	0.2262	(0.0160)
	NEV	17.2510	(2.5556)	21.1706	(3.2231)	33.8335	(1.6840)	0.2695	(0.0161)
	OII	22.0892	(5.4113)	29.6065	(6.0272)	31.7962	(1.8904)	0.2330	(0.0264)
	OS	17.8845	(0.3253)	19.8172	(2.3354)	28.2515	(0.3306)	0.2269	(0.0097)
	POP	12.7809	(6.0360)	14.9560	(6.2470)	20.6076	(6.8881)	0.2358	(0.0147)
	SRR	28.0176	(0.9656)	34.4188	(0.7000)	36.7192	(1.0015)	0.2250	(0.0038)
	WLMN	27.6775	(4.3162)	31.7387	(5.0276)	35.2975	(4.7076)	0.2024	(0.0042)
	WWW	30.9392	(1.3147)	30.1782	(4.5640)	37.6362	(0.6356)	0.2100	(0.0127)

Table 4. Likelihood Ratio Tests

In this table, we construct the likelihood ratio statistics to test if the market fundamentals have any impact on the uninformed transition probabilities and the uninformed arrival rate differential. In addition to our time-varying model (call it Model C), we also estimate two alternative models, Model A fixing the uninformed transition probabilities to be constant across time and fixing the uninformed arrival rate differential at a constant level, and Model B allowing the uninformed transition probabilities to be time-varying (dependent upon the market fundamentals) while fixing the uninformed arrival rate differential at a constant level. The likelihood ratio statistic λ_1 is constructed as twice the difference between the sample log-likelihood corresponding to the two nested Models A and C, and it is asymptotically distributed as $\chi^2(8)$ with critical value of 20.09 at the 1% level. The likelihood ratio statistic λ_2 is constructed as twice the difference between the sample log-likelihood corresponding to the two nested Models B and C, and it is asymptotically distributed as $\chi^2(4)$ with critical value of 13.28 at the 1% level.

Turnover	Ticker	λ_1	λ_2	Turnover	Ticker	λ_1	λ_2
2 nd Decile	EBF	835.39	236.58	4 th Decile	AGL	692.26	596.80
	LDL	242.13	54.61		AIZ	77.91	72.18
	OCQ	443.84	251.00		BW	814.49	57.26
	OXM	93.04	22.65		CPY	206.90	125.60
	RUS	748.55	144.16		DVI	576.52	389.84
	SL	173.50	66.24		IMC	172.80	121.96
	SMP	39.91	28.56		LUB	243.46	205.67
	STL	244.42	105.48		PWN	602.88	85.66
	TYL	286.70	106.32		RML	426.97	360.37
6 th Decile	VCD	315.79	156.45	8 th Decile	UFI	620.56	577.18
	AIR	469.25	113.62		ALN	375.70	366.71
	BDG	519.22	264.08		BGG	447.83	446.98
	ESL	1325.74	1277.89		KMT	1596.04	507.39
	FRC	658.54	501.07		NEV	189.57	182.79
	HUF	204.52	35.34		OII	370.01	311.40
	HXL	313.74	182.57		OS	290.04	282.50
	LZ	1548.07	196.63		POP	739.86	44.98
	NC	188.02	183.38		SRR	107.43	106.18
	OMI	1358.22	139.73		WLMN	267.92	85.06
	RTI	430.72	430.47		WWW	409.94	395.67

Table 5. Cross-sectional Variation of Matching Response

In this table, we present results in explaining the cross-sectional variation in the estimated probability of informed matching response, $\rho = \beta_0 + \beta_1 \cdot \text{INVOL} + \beta_2 \cdot \text{UNVOL} + \varepsilon$. The dependent variable ρ is the estimated probability of informed matching response; the implied dollar volume by informed traders INVOL is defined as the product of mean probability of informed trading $\overline{\text{TPIN}}$ and the mean dollar volume $\overline{\text{VOL}}$; and the implied dollar volume by uninformed traders UNVOL is defined as the product of $1 - \overline{\text{TPIN}}$ and the mean dollar volume $\overline{\text{VOL}}$. The dollar volume VOL is defined as the product of stock price and share volume (in thousand shares). Both means are the sample average of daily data between January 1993 and December 2002. The sample size is 40, and the t-statistics are reported in parentheses.

Intercept	INVOL	UNVOL	Adj. R ²	F-Stat
0.9442	-6.93E-04	1.43E-04	0.4353	16.03
(65.21)	(-4.23)	(3.35)		

Table 6. Information Content and Stock Spreads

In this table, we present results for a fixed-effect panel regression with modification to the original Easley et al. (1996) design, $\Sigma_{i,t} = \beta_{0,i} + \beta_1 \cdot \text{VTPIN}_{i,t} + \beta_2 \cdot \text{VPIN}_{i,t} + \beta_3 \cdot \text{VOL}_{i,t} + \beta_4 \cdot \delta_2(t) + \beta_5 \cdot \delta_3(t) + \beta_6 \cdot \delta_2(t) \cdot \text{VTPIN}_{i,t} + \beta_7 \cdot \delta_2(t) \cdot \text{VPIN}_{i,t} + \beta_8 \cdot \delta_3(t) \cdot \text{VTPIN}_{i,t} + \beta_9 \cdot \delta_3(t) \cdot \text{VPIN}_{i,t} + \eta_{i,t}$. Σ is the stock spread of different types extracted from the NYSE TAQ database. VPIN is the product of stock price and probability of informed trading PIN estimated from the Easley et al. (1996) framework. VTPIN is the product of stock price and estimated probability of informed trading TPIN from our extended model. VOL is the dollar volume defined as the product of stock price and share volume (in thousand shares). $\delta_2(t)$ is a dummy variable that takes the value 1 for the period of tick size being 1/16 on the NYSE (i.e., between June 24, 1997 and January 28, 2001). $\delta_3(t)$ is a dummy variable that takes the value 1 for the period of post-decimalization on the NYSE (i.e., on and after January 29, 2001). Panel (A) uses 99379 daily observations between January 1993 and December 2002, and Panel (B) uses 1597 quarterly observations for the same period. The t-statistics are reported in parentheses.

Panel (A): Contemporaneous Relationship (daily)						
Regressors	Opening Spread		Closing Spread		Mean Spread	
VTPIN	0.012	(9.01)	0.008	(26.88)	0.008	(48.04)
VPIN	0.000	(-0.02)	0.003	(7.47)	0.002	(13.34)
VOL	-9.59E-07	(-1.73)	-9.90E-07	(-7.40)	-8.39E-07	(-11.84)
δ_2	-0.039	(-7.42)	-0.032	(-25.01)	-0.039	(-56.98)
δ_3	-0.155	(-24.00)	-0.085	(-54.35)	-0.096	(-116.02)
$\delta_2 \cdot \text{VTPIN}$	-0.004	(-2.37)	-0.006	(-13.32)	-0.005	(-20.19)
$\delta_2 \cdot \text{VPIN}$	0.007	(3.28)	0.003	(5.35)	0.003	(11.86)
$\delta_3 \cdot \text{VTPIN}$	0.000	(-0.17)	-0.007	(-10.72)	-0.006	(-17.42)
$\delta_3 \cdot \text{VPIN}$	0.027	(7.99)	0.003	(3.52)	0.003	(6.30)

Panel (B): Contemporaneous Relationship (quarterly)						
Regressors	Opening Spread		Closing Spread		Mean Spread	
VTPIN	0.014	(8.45)	0.011	(13.63)	0.010	(13.35)
VPIN	-0.001	(-0.62)	0.002	(2.16)	0.002	(2.13)
VOL	-9.37E-06	(-5.28)	-8.17E-06	(-9.63)	-7.50E-06	(-9.27)
δ_2	-0.039	(-5.96)	-0.032	(-10.08)	-0.038	(-12.69)
δ_3	-0.150	(-18.98)	-0.084	(-22.14)	-0.095	(-26.24)
$\delta_2 \cdot \text{VTPIN}$	-0.004	(-1.57)	-0.004	(-4.14)	-0.004	(-3.41)
$\delta_2 \cdot \text{VPIN}$	0.007	(2.56)	0.002	(1.36)	0.002	(1.90)
$\delta_3 \cdot \text{VTPIN}$	0.002	(0.54)	-0.005	(-2.72)	-0.004	(-2.34)
$\delta_3 \cdot \text{VPIN}$	0.024	(5.73)	0.001	(0.44)	0.001	(0.50)

Table 7. Competing Measures of Information Asymmetry

In this table, we report results for predicting the mean bid-ask spread on the next trading period using a set of competing measures of information asymmetry in a fixed-effect panel regression framework. The regression design is $\Sigma_{i,t+1} = \beta_{0,i} + \beta_1 \cdot \text{VTPIN}_{i,t} + \beta_2 \cdot \text{VPIN}_{i,t} + \beta_3 \cdot \text{VOL}_{i,t} + \beta_4 \cdot \text{AMS}_{i,t} + \beta_5 \cdot \text{AVOL}_{i,t} + \beta_6 \cdot \text{OIMB}_{i,t} + \beta_7 \cdot \text{ME}_{i,t} + \beta_8 \cdot \text{RVOL}_{i,t} + \beta_9 \cdot \delta_2(t) + \beta_{10} \cdot \delta_3(t) + \beta_{11} \cdot \delta_2(t) \cdot \text{VTPIN}_{i,t} + \beta_{12} \cdot \delta_2(t) \cdot \text{VPIN}_{i,t} + \beta_{13} \cdot \delta_3(t) \cdot \text{VTPIN}_{i,t} + \beta_{14} \cdot \delta_3(t) \cdot \text{VPIN}_{i,t} + \eta_{i,t+1}$. Σ is the mean bid-ask spread computed from the NYSE TAQ database. VPIN is the product of stock price and probability of informed trading PIN estimated from the Easley et al. (1996) framework. VTPIN is the product of stock price and estimated probability of informed trading TPIN from our extended model. VOL is the dollar volume defined as the product of stock price and share volume (in thousand shares). AMS is the abnormal mean spread computed as the deviation of current mean bid-ask spread from the moving average of past 20-trading-day mean bid-ask spread. AVOL is the abnormal volume computed as the deviation of current dollar volume from the moving average of past 20-trading-day mean dollar volume. OIMB is the order imbalance or absolute net order flow in number of trades. ME is the market value of equity. RVOL is the volatility of returns in the past 20-trading-day period. $\delta_2(t)$ is a dummy variable that takes the value 1 for the period of tick size being 1/16 on the NYSE (i.e., between June 24, 1997 and January 28, 2001). $\delta_3(t)$ is a dummy variable that takes the value 1 for the period of post-decimalization on the NYSE (i.e., on and after January 29, 2001). Panel (A) uses 98638 daily observations between January 1993 and December 2002, and Panel (B) uses 1557 quarterly observations for the same period. The t-statistics are reported in parentheses.

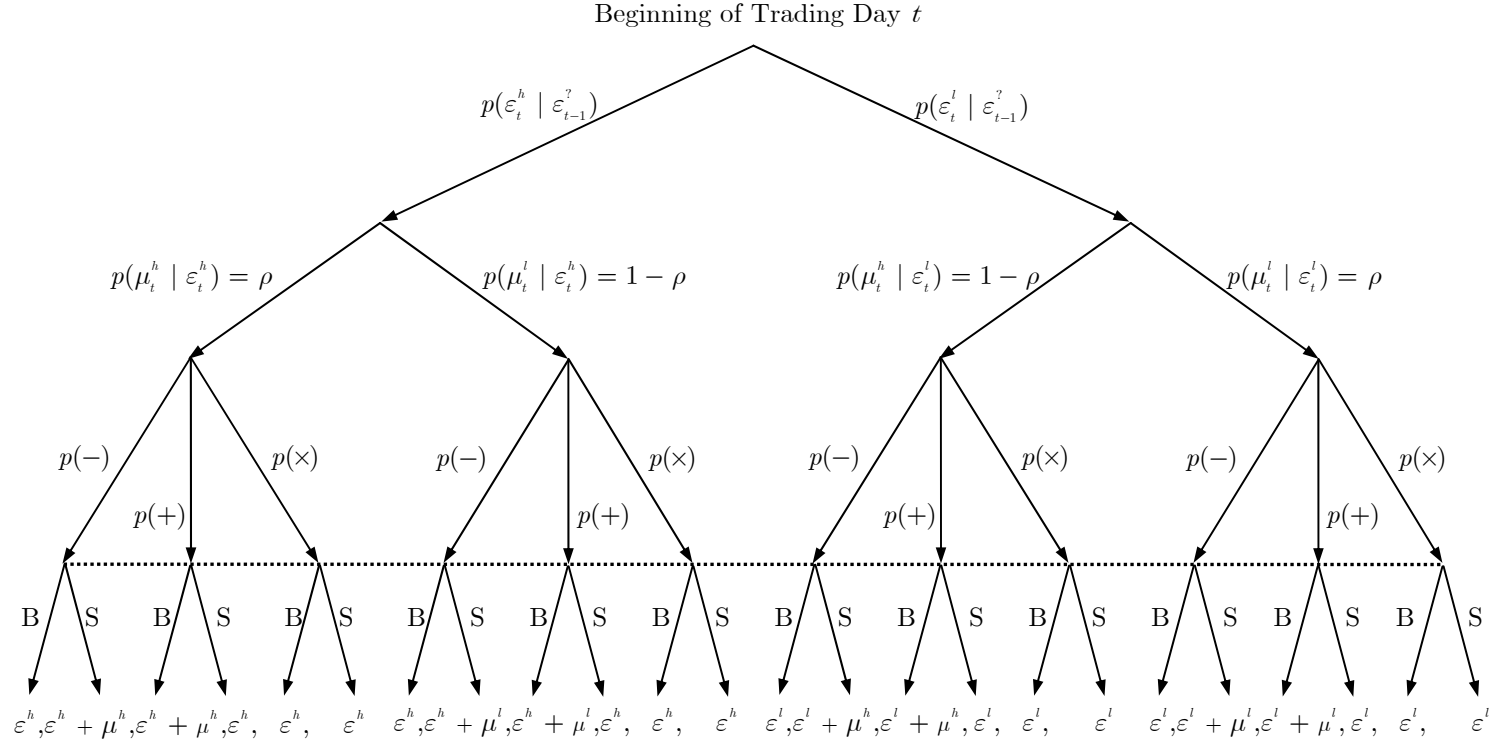
Regressors	Panel (A)		Panel (B)	
VTPIN	0.011	(57.29)	0.009	(10.78)
VPIN	0.002	(11.55)	0.004	(4.67)
VOL	-4.45E-06	(-27.65)	-2.00E-06	(-1.91)
AMS	0.172	(47.86)	0.811	(5.77)
AVOL	4.11E-06	(25.09)	-1.27E-06	(-0.37)
OIMB	-2.93E-04	(-20.02)	-1.98E-03	(-9.28)
ME	-1.50E-08	(-13.04)	-2.44E-08	(-4.33)
RVOL	0.402	(27.67)	0.328	(3.39)
δ_2	-0.041	(-60.37)	-0.036	(-11.80)
δ_3	-0.099	(-118.21)	-0.085	(-20.00)
$\delta_2 \cdot \text{VTPIN}$	-0.004	(-18.62)	-0.003	(-3.20)
$\delta_2 \cdot \text{VPIN}$	0.003	(11.37)	0.003	(2.00)
$\delta_3 \cdot \text{VTPIN}$	-0.005	(-13.73)	0.001	(0.41)
$\delta_3 \cdot \text{VPIN}$	0.002	(5.85)	-0.003	(-1.73)

Table 8. Information Asymmetry and Autocorrelation Returns

In this table, we present results for the following two fixed-effect panel regression models, $r_{i,t+1} = \gamma_{0,i} + r_{i,t}(\gamma_1 + \gamma_2 \cdot ia_{i,t}) + \varepsilon_{i,t+1}$, and $r_{i,t+1} = \gamma_{0,i} + r_{i,t} \cdot [\gamma_1 + (\gamma_{2,2} + \gamma_{2,4} \cdot \tau_4 + \gamma_{2,6} \cdot \tau_6 + \gamma_{2,8} \cdot \tau_8) \cdot ia_{i,t}] + \varepsilon_{i,t+1}$. The independent variable r is the daily holding period stock return from CRSP. The dependent variable ia is one of the following three measures of information asymmetry: VOL is the dollar volume defined as the product of stock price and share volume (in thousand shares), MSPR is the daily mean bid-ask spread extracted from NYSE TAQ, and VTPIN is the product of stock price and estimated probability of informed trading from our model as an extension of Easley et al. (1996). The turnover group dummies τ_4 , τ_6 and τ_8 take the value 1 for stocks in the 4th, 6th and 8th turnover decile, respectively. There are 99551 daily observations for the panel regression during the period between January 1993 and December 2002, and the t-statistics are reported in parentheses.

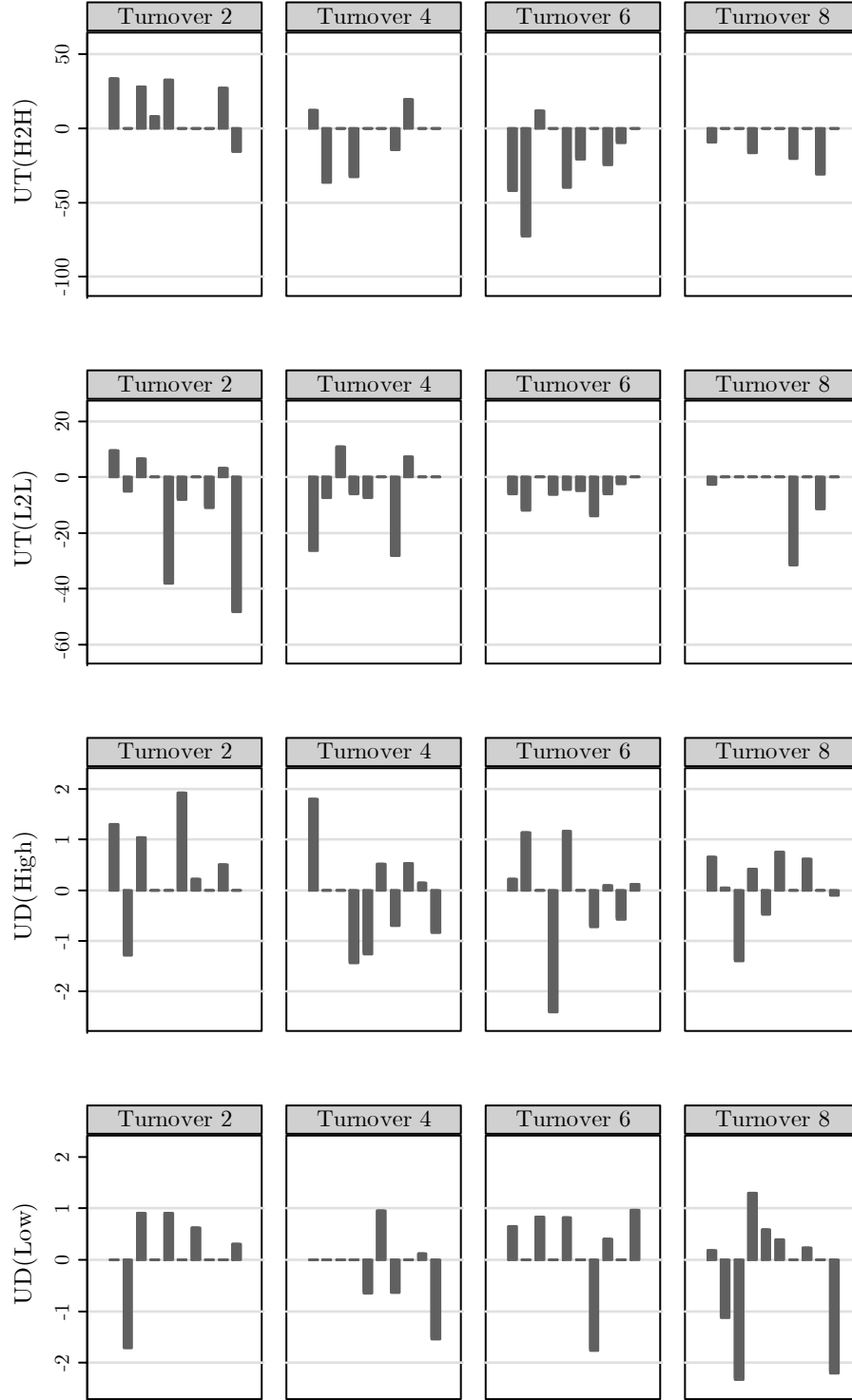
ia	r	r · ia	r · ia · τ_4	r · ia · τ_6	r · ia · τ_8
VOL	-0.0162 (-4.62)	2.27E-07 (0.34)			
VOL	-0.0160 (-4.45)	3.16E-07 (0.07)	6.33E-07 (0.14)	-6.2E-07 (-0.14)	-2.9E-07 (-0.06)
MSPR	-0.0179 (-2.30)	0.0140 (0.31)			
MSPR	-0.0218 (-2.76)	-0.1315 (-2.51)	0.2090 (3.89)	0.2329 (4.82)	0.2621 (4.97)
VTPIN	-0.0506 (-9.85)	0.0105 (8.63)			
VTPIN	-0.0553 (-10.05)	0.0172 (6.56)	-0.0045 (-1.56)	-0.0069 (-2.71)	-0.0073 (-2.85)

Figure 1. Structure of Trading Process on a Typical Day



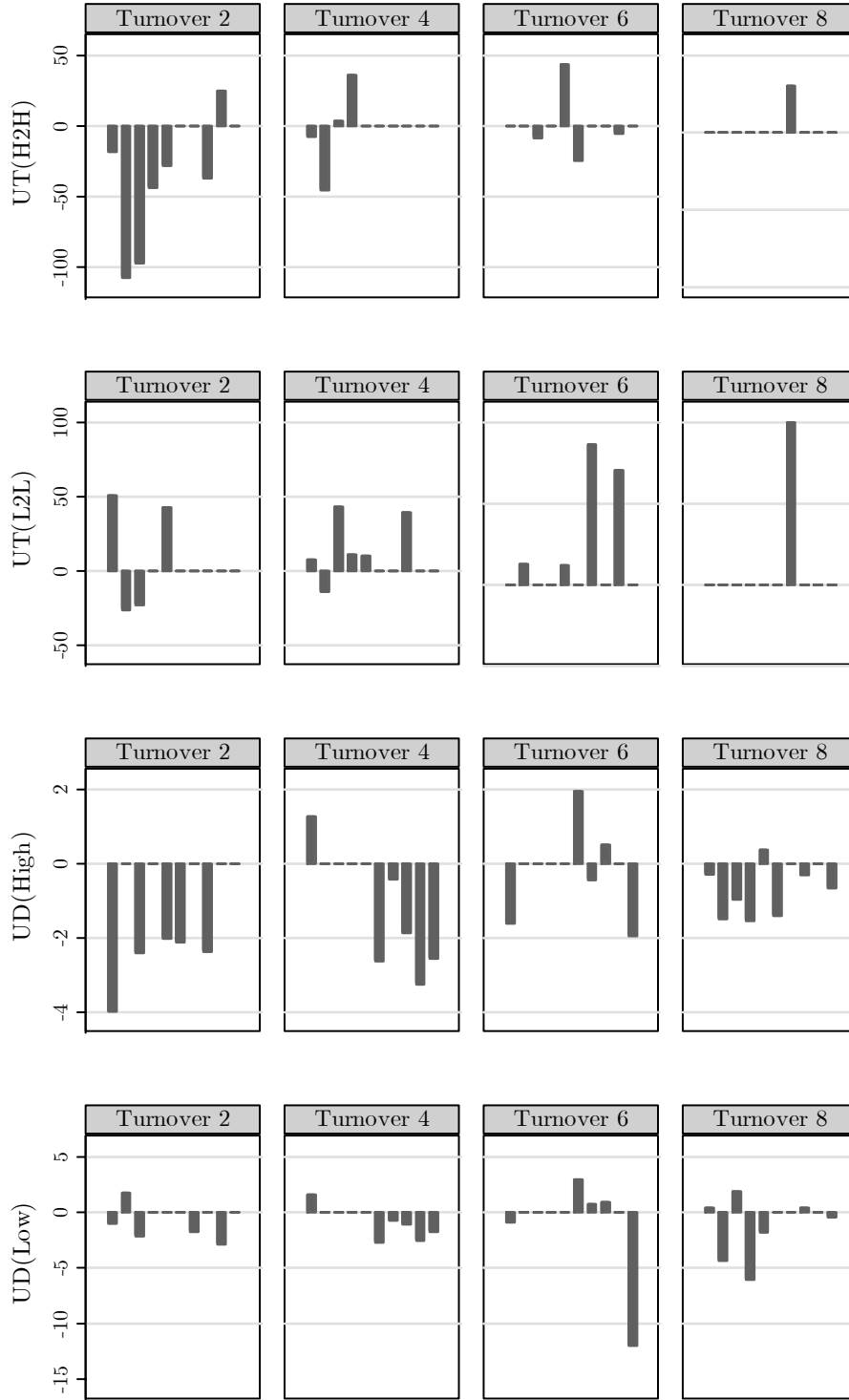
In the figure above, ε^h and ε^l are the high and low levels of the uninformed arrival rate, μ^h and μ^l are the high and low levels of the informed arrival rate. Depending upon the arrival rate level $\varepsilon_{t-1}^?$ in the previous day, the uninformed arrival may transit into the high state with probability $p(\varepsilon_t^h | \varepsilon_{t-1}^?)$ or the low state with probability $p(\varepsilon_t^l | \varepsilon_{t-1}^?)$. The informed traders engage in level-matching activities with probability ρ . There may be no news arrival with probability $p(\times)$, good news arrival with probability $p(+)$, or bad news with probability $p(-)$. Nodes above the dotted line occur only once per trading day and nodes on the dotted line are repeated many times each day. The buy and sell orders are indicated by B and S, respectively. The last row indicates the aggregate, expected arrival rate of traders under each scenario, with the buy and sell nature for ε^h and ε^l abbreviated.

Figure 2. Influence of Prior Stock Return (CDR)



In the figure above, CDR is the cumulative prior-20-trading-day stock return net three-month T-bill rate; UT(H2H) and UT(L2L) are the estimated coefficients associated with CDR on the uninformed transition probability remaining at the high and the low state, respectively; and UD(High) and UD(Low) are the estimated coefficients associated with CDR on the uninformed arrival rate differential at the high and the low state, respectively. We assign zero value to the estimated coefficients that are not statistically significantly different from zero at the 5% level. The columns correspond to each of the 2nd, 4th, 6th, and 8th turnover deciles.

Figure 3. Influence of Prior Market Return (CMR)



In the figure above, CMR is the cumulative prior-20-trading-day market return net of three-month T-bill rate; UT(H2H) and UT(L2L) are the estimated coefficients associated with CMR on the uninformed transition probability remaining at the high and the low state, respectively; and UD(High) and UD(Low) are the estimated coefficients associated with CMR on the uninformed arrival rate differential at the high and the low state, respectively. We assign zero value to the estimated coefficients that are not statistically significantly different from zero at the 5% level. The columns correspond to each of the 2nd, 4th, 6th, and 8th turnover deciles.